THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

# ISE 430/530: Manufacturing Systems and Facilities Design

Dr. Cheng (Erick) Chen
Industrial & Systems and Engineering Management Department

Cheng.chen@uah.edu

# Using Excel Solver

## Review: Linear Programming Using Excel Solver

$$Maximize\ Z = x_1 + 2x_2$$

Subject to

$$x_1 + x_2 \leq 10$$
$$x_1 + 3x_2 \leq 20$$
$$x_1, x_2 \geq 0$$

$$Maximize\ Z = \underbrace{[1, 2]}_{\boldsymbol{C^T}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{X}}$$

Subject to

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}}_{\boldsymbol{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{X}} \leq \underbrace{\begin{bmatrix} 10 \\ 20 \end{bmatrix}}_{\boldsymbol{B}}$$

$$x_1, x_2 \geq 0$$

$$(1, n)(n, 1) = 1$$

$$Maximize\ Z = C^T X$$

Subject to

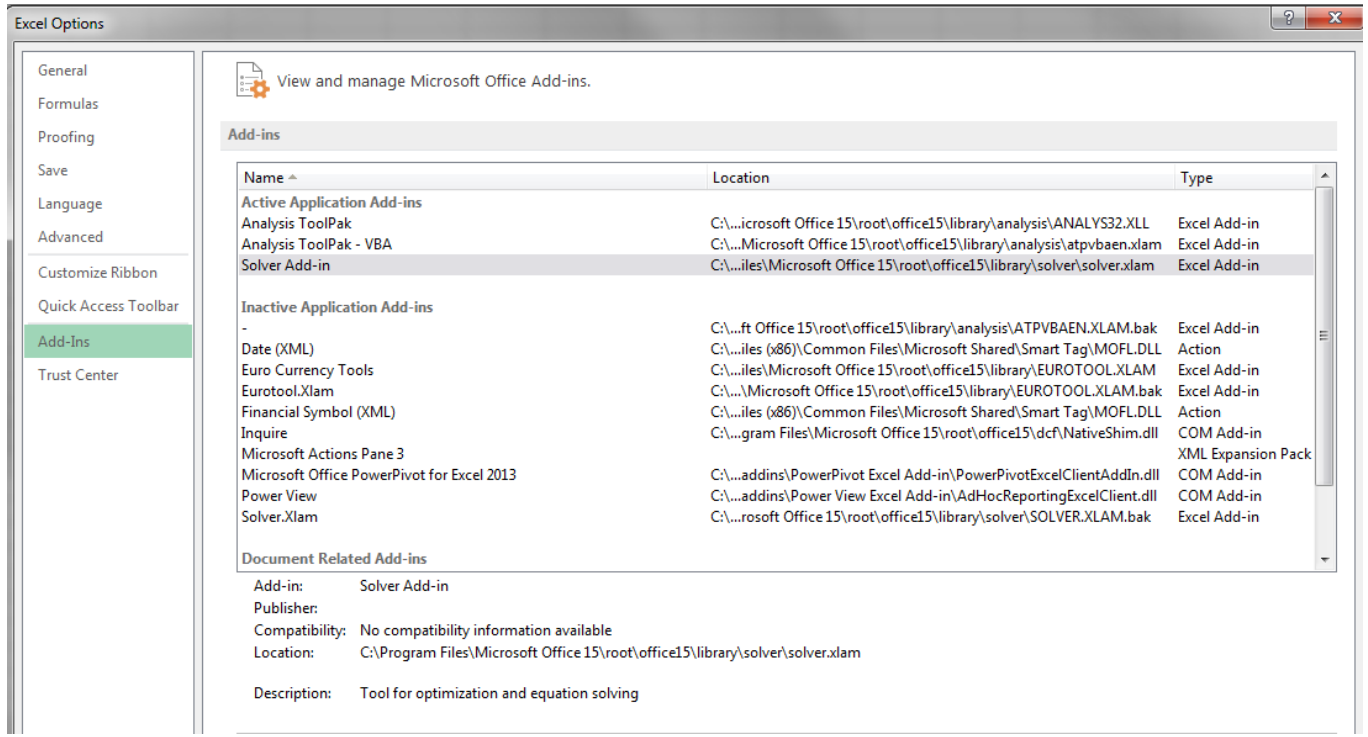
$$AX \leq B$$

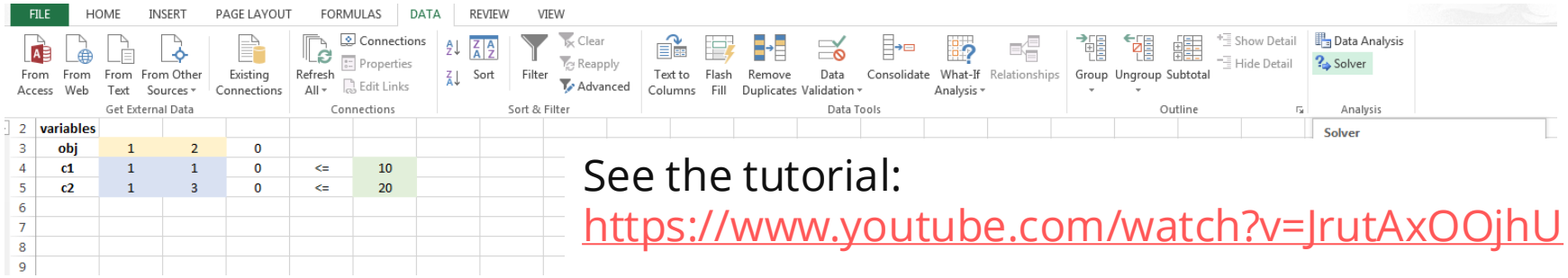
$$(m, n)(n, 1) = (m, 1)$$

C:n by 1 matrix
X:n by 1 matrix
A: m by n matrix
$B$: m by 1 matrix

n = number of variables
m = number of constraints

# Using Excel Solver




See the tutorial:
https://www.youtube.com/watch?v=JrutAxOOjhU

# Using Excel Solver

## Review: Linear Programming Using Excel Solver

$$Maximize\ Z = \underbrace{[1, 2]}_{C^T} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{X}$$

Subject to

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{X} \leq \underbrace{\begin{bmatrix} 10 \\ 20 \end{bmatrix}}_{B}$$

$$x_1, x_2 \geq 0$$

$$Maximize\ Z = C^T X$$

Subject to

$$AX \leq B$$

| | x1 | x2 | | | |
|---|---|---|---|---|---|
| variables | | | $X$ | | |
| obj | 1 | 2 | $C^T$ 0 | | $B$ |
| c1 | 1 | 1 | 0 | <= | 10 |
| c2 | 1 | 3 | 0 | <= | 20 |
| | | $A$ | | | |

=sumproduct( , )

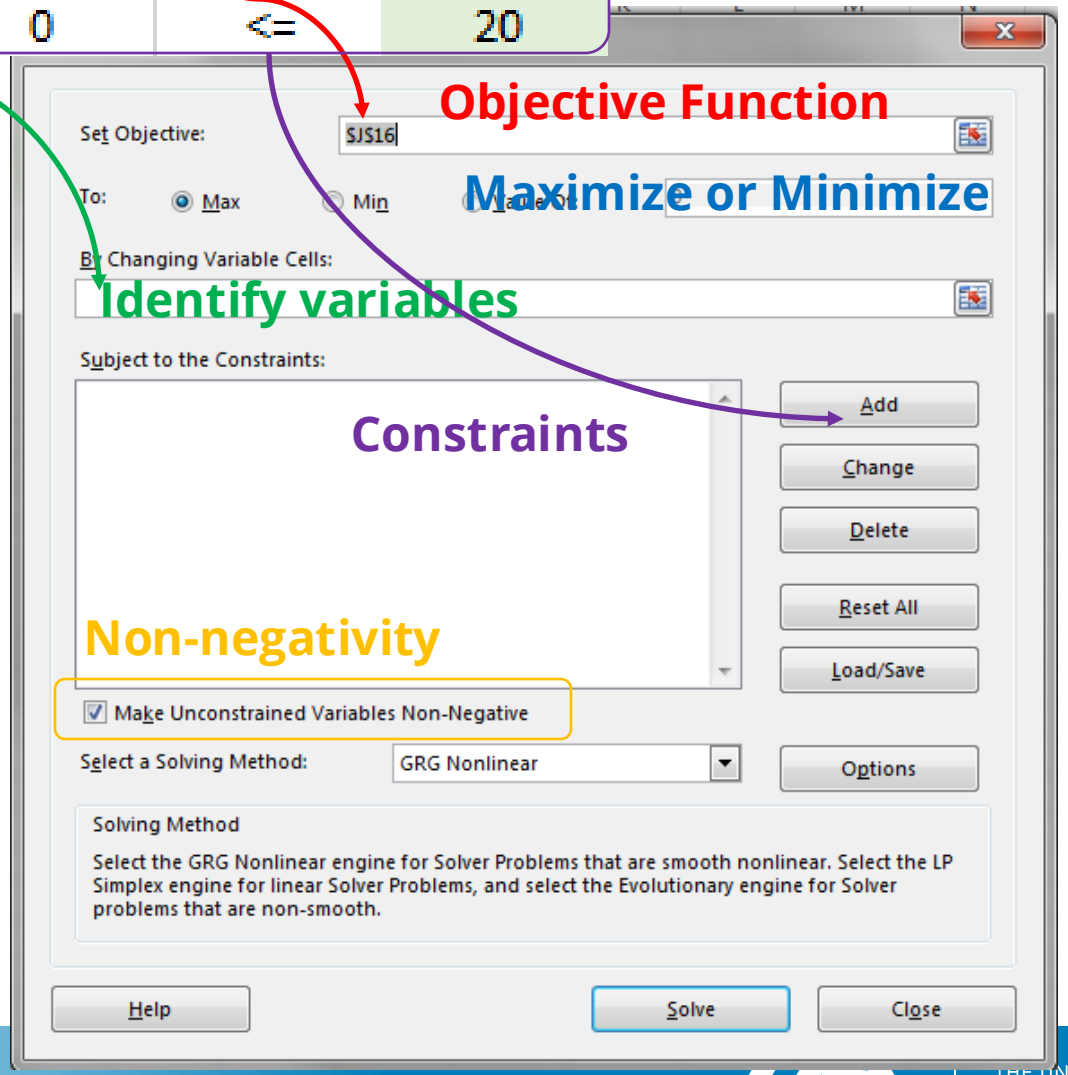
=sumproduct( , )

=sumproduct( , )



THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

| | x1 | x2 | | | |
|---|---|---|---|---|---|
| variables | | | | | |
| obj | 1 | 2 | 0 | | |
| c1 | 1 | 1 | 0 | <= | 10 |
| c2 | 1 | 3 | 0 | <= | 20 |



| | x1 | x2 | | | |
|---|---|---|---|---|---|
| variables | | | | | |
| obj | 1 | 2 | 0 | | |
| c1 | 1 | 1 | 0 | <= | 10 |
| c2 | 1 | 3 | 0 | <= | 20 |

Microsoft Excel 15.0 Answer Report
Worksheet: [Book1]Sheet1
Report Created: 3/10/2014 10:27:31 PM
Result: Solver found a solution. All Constraints and optimality conditions are satisfied.
Solver Engine
Engine: GRG Nonlinear
Solution Time: 0 Seconds.
Iterations: 3 Subproblems: 0
Solver Options
Max Time Unlimited, Iterations Unlimited, Precision 0.000001
Convergence 0.0001, Population Size 100, Random Seed 0, Derivatives Central
Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Max)

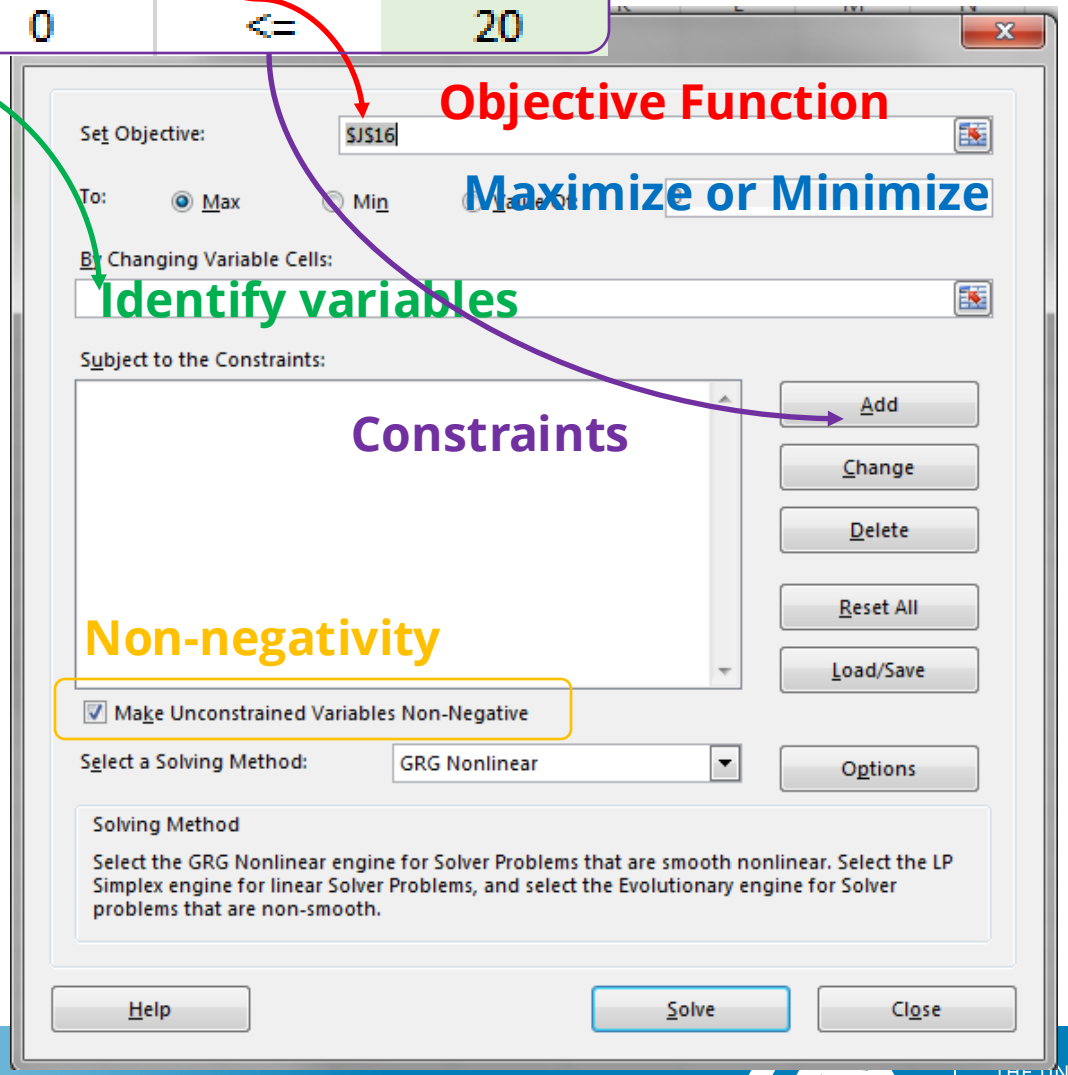
| Cell | Name | Original Value | Final Value |
|---|---|---|---|
| $D$3 | obj | 0 | 15 |

Variable Cells

| Cell | Name | Original Value | Final Value | Integer |
|---|---|---|---|---|
| $B$2 | variables x1 | 0 | 5 | Contin |
| $C$2 | variables x2 | 0 | 5 | Contin |

Constraints

| Cell | Name | Cel | Value | Formula | Status | Slack |
|---|---|---|---|---|---|---|
| $D$4 | c1 | | 10 | $D$4<=$F$4 | Binding | 0 |
| $D$5 | c2 | | 20 | $D$5<=$F$5 | Binding | 0 |



$$Minimize\ 10(\alpha_{12}^{+}+\alpha_{12}^{-}+\beta_{12}^{+}+\beta_{12}^{-})$$
$$+20(\alpha_{13}^{+}+\alpha_{13}^{-}+\beta_{13}^{+}+\beta_{13}^{-})+50(\alpha_{23}^{+}+\alpha_{23}^{-}+\beta_{23}^{+}+\beta_{23}^{-})$$

$$37.5 \le x_1'' - x_1',\ x_1'' - x_1' \le 100$$
$$12.5 \le x_2'' - x_2',\ x_2'' - x_2' \le 40$$
$$50 \le x_3'' - x_3',\ x_3'' - x_3' \le 100$$
$$30 \le y_1'' - y_1',\ y_1'' - y_1' \le 80$$
$$10 \le y_2'' - y_2',\ y_2'' - y_2' \le 80$$
$$40 \le y_3'' - y_3',\ y_3'' - y_3' \le 80$$
$$109 \le (x_1'' - x_1' + y_1'' - y_1') \le 130$$
$$63 \le (x_2'' - x_2' + y_2'' - y_2') \le 110$$
$$126 \le (x_3'' - x_3' + y_3'' - y_3') \le 140$$

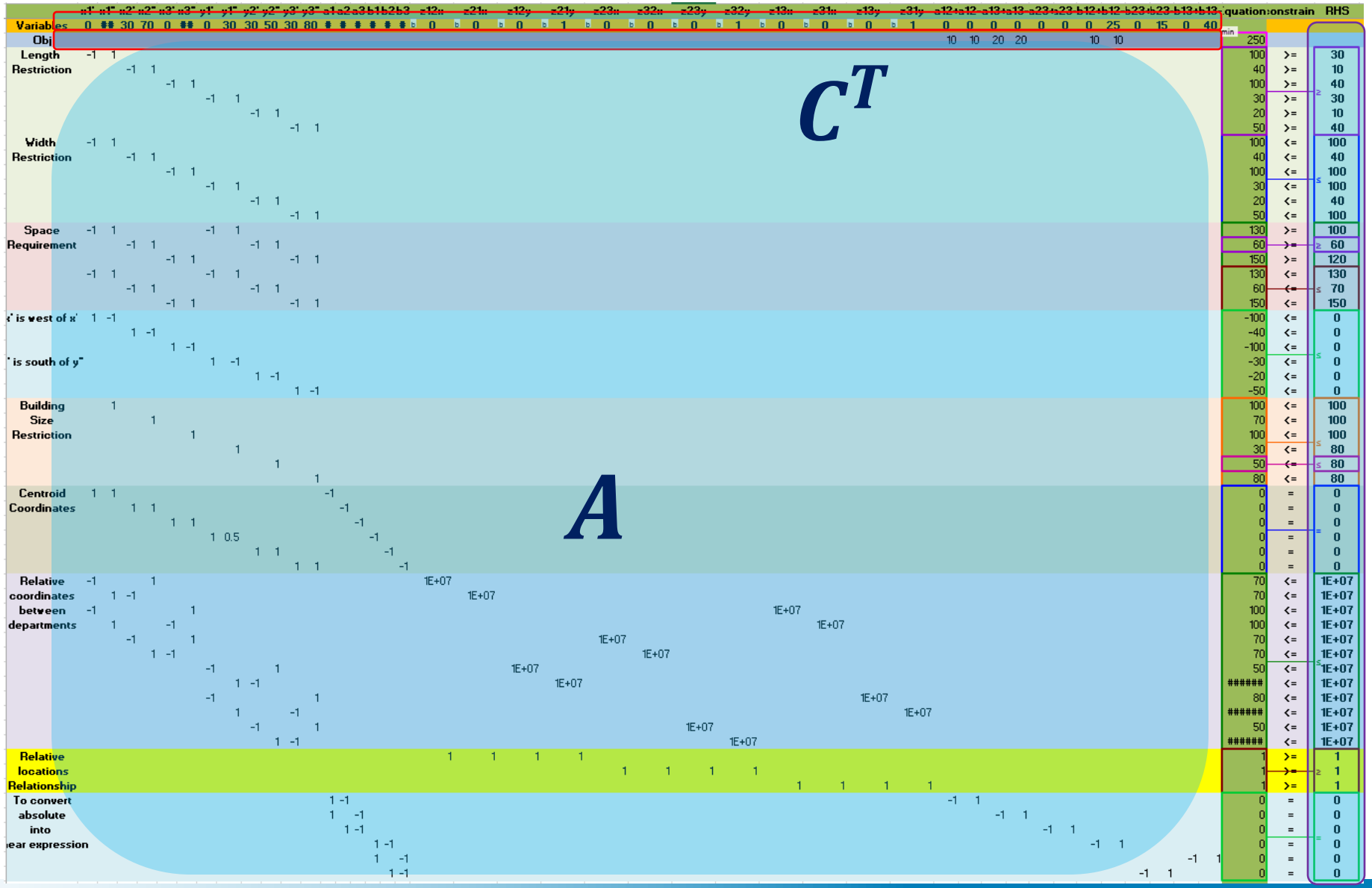
$$x_1' \le x_1'' \quad y_1' \le y_1'' \quad \alpha_1 = 0.5x_1' + 0.5x_1''$$
$$x_2' \le x_2'' \quad y_2' \le y_2'' \quad \alpha_2 = 0.5x_2' + 0.5x_2''$$
$$x_3' \le x_3'' \quad y_3' \le y_3'' \quad \alpha_3 = 0.5x_3' + 0.5x_3''$$
$$x_1'' \le 100 \quad y_1'' \le 80 \quad \beta_1 = 0.5y_1' + 0.5y_1''$$
$$x_2'' \le 100 \quad y_1'' \le 80 \quad \beta_2 = 0.5y_2' + 0.5y_2''$$
$$x_3'' \le 100 \quad y_1'' \le 80 \quad \beta_3 = 0.5y_3' + 0.5y_3''$$

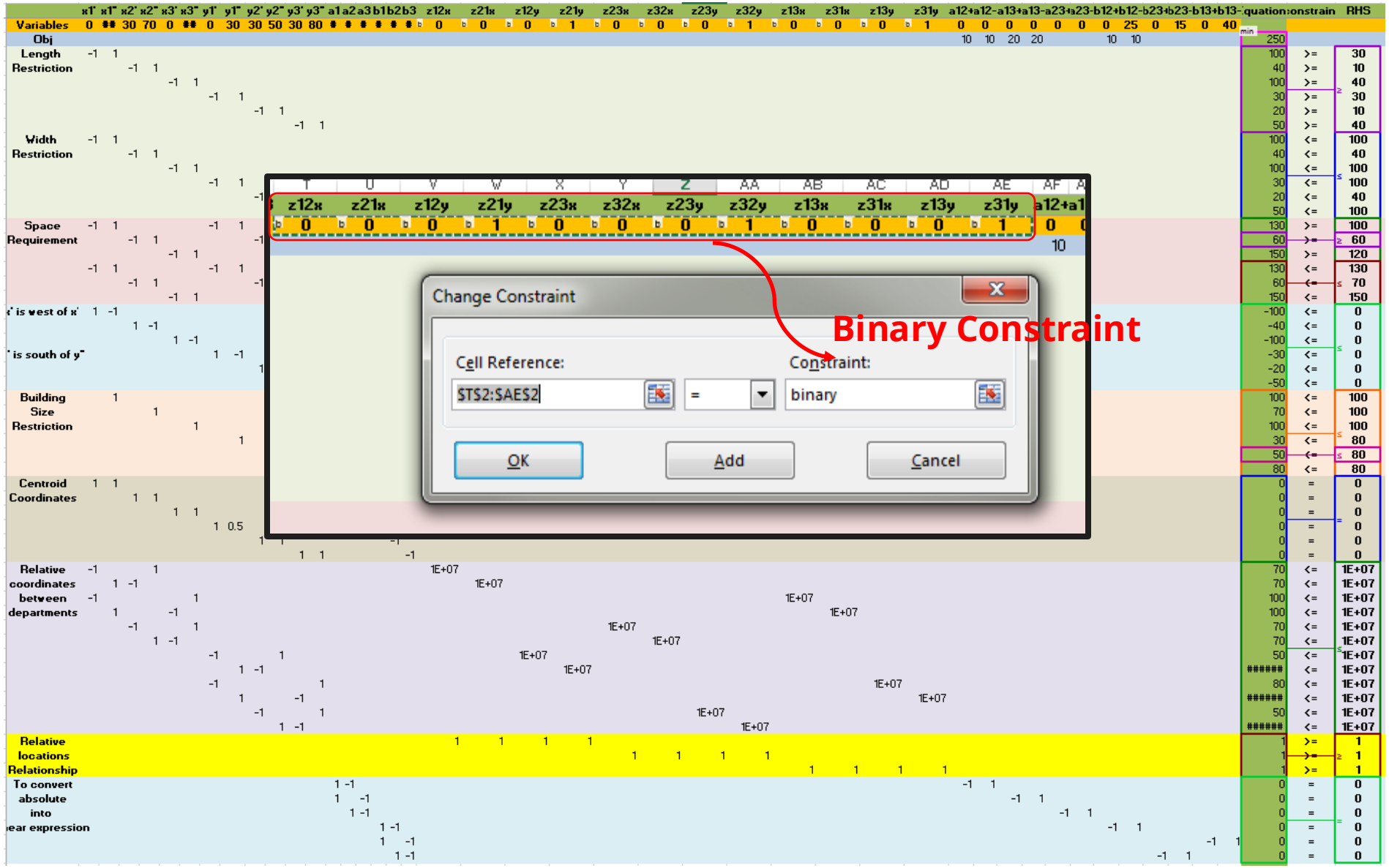
$$x_2'' \le x_1' + (1 - z_{12}^{x})M \quad y_2'' \le y_1' + (1 - z_{12}^{y})M$$
$$x_1'' \le x_2' + (1 - z_{21}^{x})M \quad y_1'' \le y_2' + (1 - z_{21}^{y})M$$
$$x_3'' \le x_1' + (1 - z_{13}^{x})M \quad y_3'' \le y_1' + (1 - z_{13}^{y})M$$
$$x_1'' \le x_3' + (1 - z_{31}^{x})M \quad y_1'' \le y_3' + (1 - z_{31}^{y})M$$
$$x_3'' \le x_1' + (1 - z_{13}^{x})M \quad y_3'' \le y_2' + (1 - z_{23}^{y})M$$
$$x_1'' \le x_3' + (1 - z_{31}^{x})M \quad y_2'' \le y_3' + (1 - z_{32}^{y})M$$

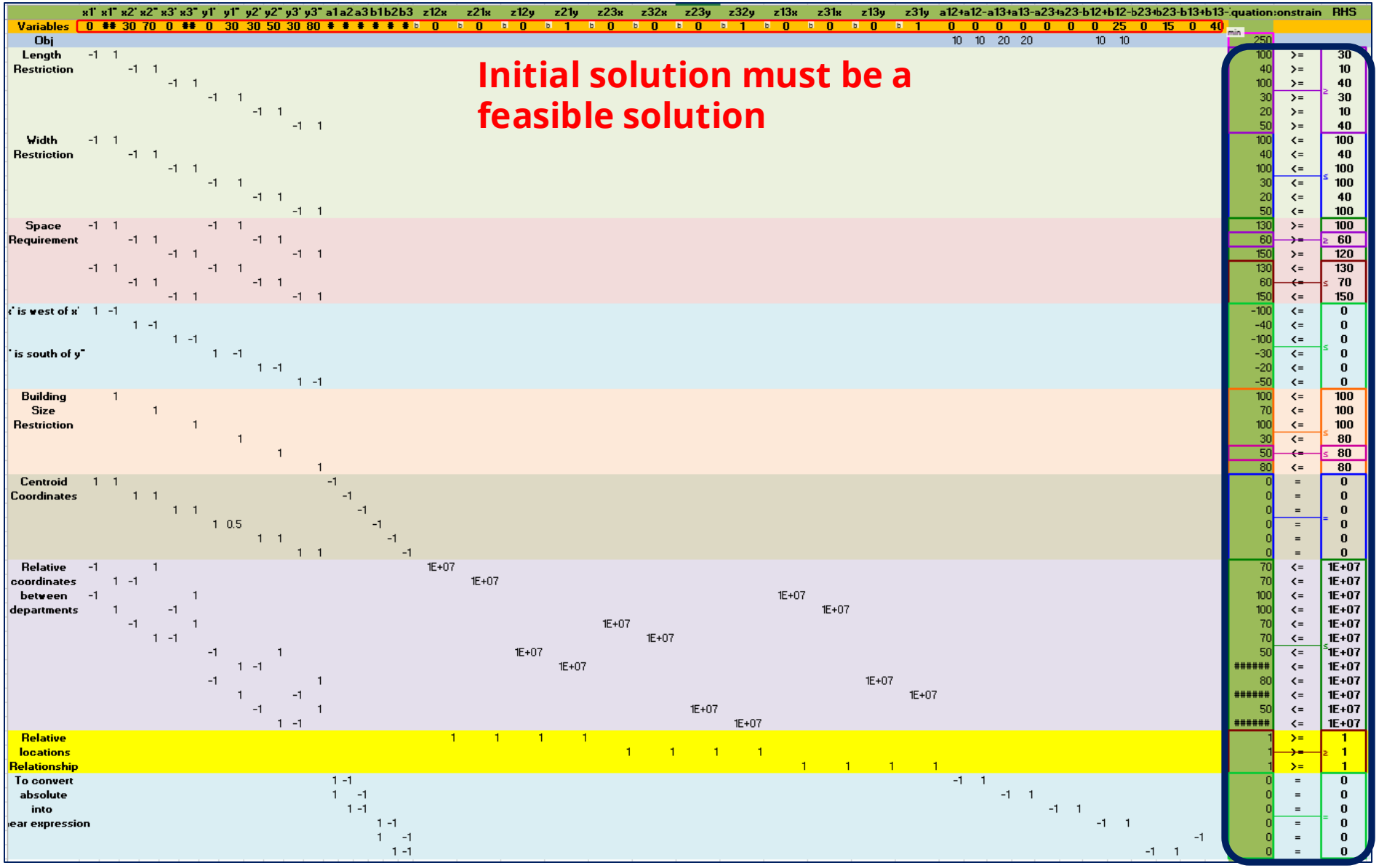
$$z_{12}^{x} + z_{21}^{x} + z_{12}^{y} + z_{21}^{y} + \ge 1$$
$$z_{13}^{x} + z_{31}^{x} + z_{13}^{y} + z_{31}^{y} + \ge 1$$
$$z_{23}^{x} + z_{32}^{x} + z_{23}^{y} + z_{32}^{y} + \ge 1$$

$$\alpha_1 - \alpha_2 = \alpha_{12}^{+} - \alpha_{12}^{-}$$
$$\alpha_1 - \alpha_3 = \alpha_{13}^{+} - \alpha_{13}^{-}$$
$$\alpha_2 - \alpha_3 = \alpha_{23}^{+} - \alpha_{23}^{-}$$
$$\beta_1 - \beta_2 = \beta_{12}^{+} - \beta_{12}^{-}$$
$$\beta_1 - \beta_3 = \beta_{13}^{+} - \beta_{13}^{-}$$
$$\beta_2 - \beta_3 = \beta_{23}^{+} - \beta_{23}^{-}$$

$$\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3 \ge 0$$
$$x_1', x_1'', y'_1, y_1'', x_2', x_2'', y'_2, y_2'', x_2', x_2'', y'_2, y_2'' \ge 0$$
$$z_{12}^{x}, z_{12}^{y}, z_{21}^{x}, z_{21}^{y}, z_{13}^{x}, z_{13}^{y}, z_{31}^{x}, z_{31}^{y}, z_{23}^{x}, z_{23}^{y}, z_{32}^{x}, z_{32}^{y} = \{0,1\}$$


X
B
$C^T$
A
Variables
Obj
Length Restriction
Width Restriction
Space Requirement
Building Size Restriction
Centroid Coordinates
Relative coordinates between departments
Relative locations Relationship
To convert absolute into linear expression
RHS


Change Constraint
Cell Reference:
$T$2:$AE$2
=
Constraint:
binary
OK
Add
Cancel
Binary Constraint


Initial solution must be a feasible solution

- Please practice on Excel for MIP based on the MIP problem we had

# Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

# LOGIC

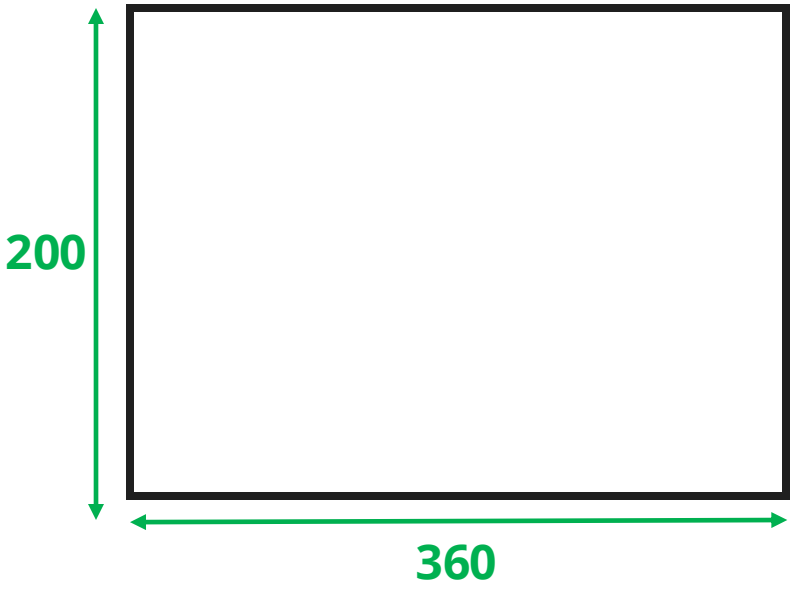
## Layout Optimization with Guillotine Induced Cuts

- Inputs are from-to chart, total area, department areas
- Distance-based objective function
- Improvement algorithm
- Divides building into smaller and smaller sections by successive horizontal or vertical cuts
- Stops when each section is a single department
- Constructs a tree to keep track of cuts and subsections
- Choice of cuts made using search strategy (i.e. simulated annealing)

# Example 6.6

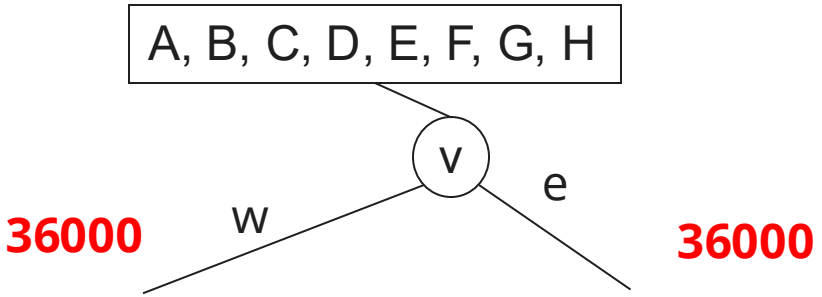
A manufacturing facility with 7 departments. Total available space is 72,000 $ft^2$. We use 20x20 grid as the unit area. For practical reasons, locations of receiving **(A)** and shipping **(G)** departments are fixed.

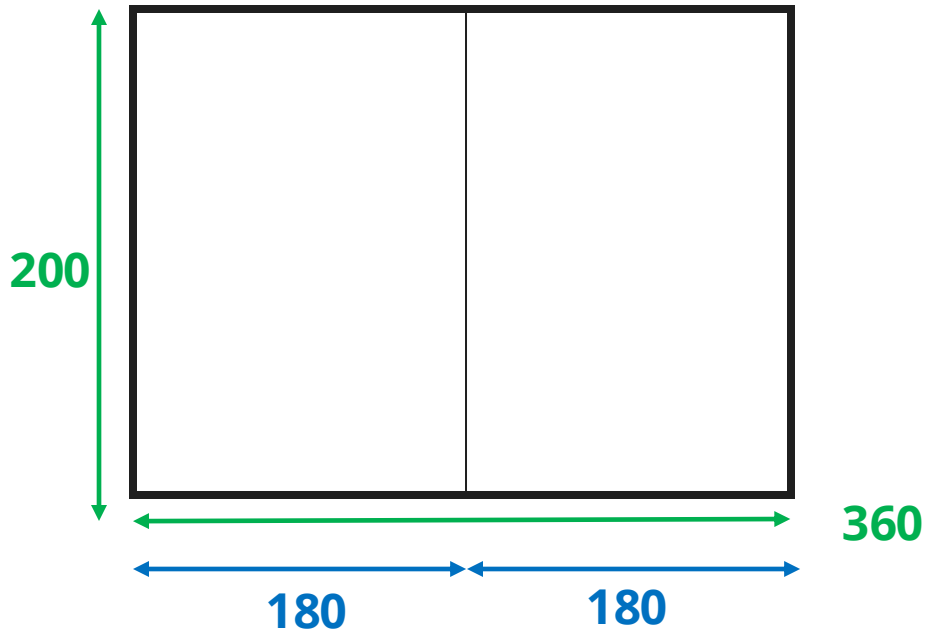
From-to chart and building dimensions (360 X 200) are given.




| | | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Department Name | Area ($ft^2$) | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

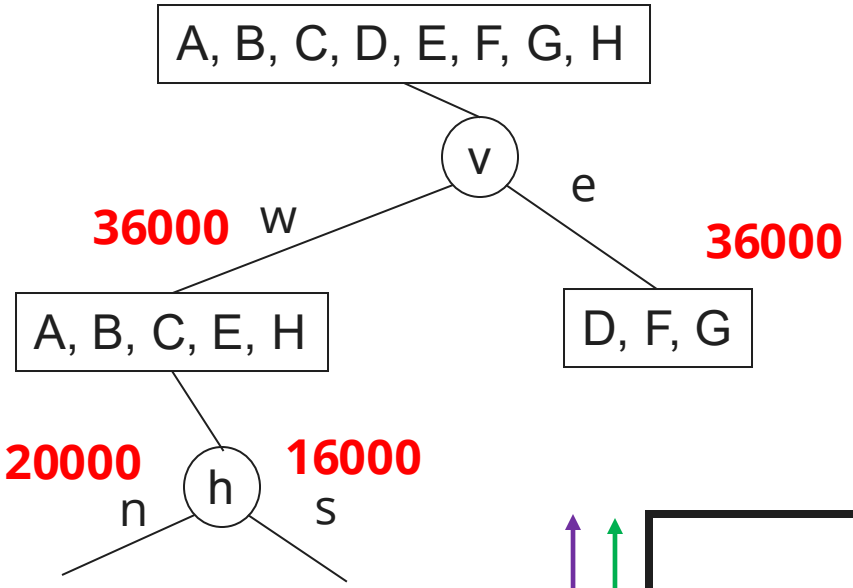
# Example 6.6



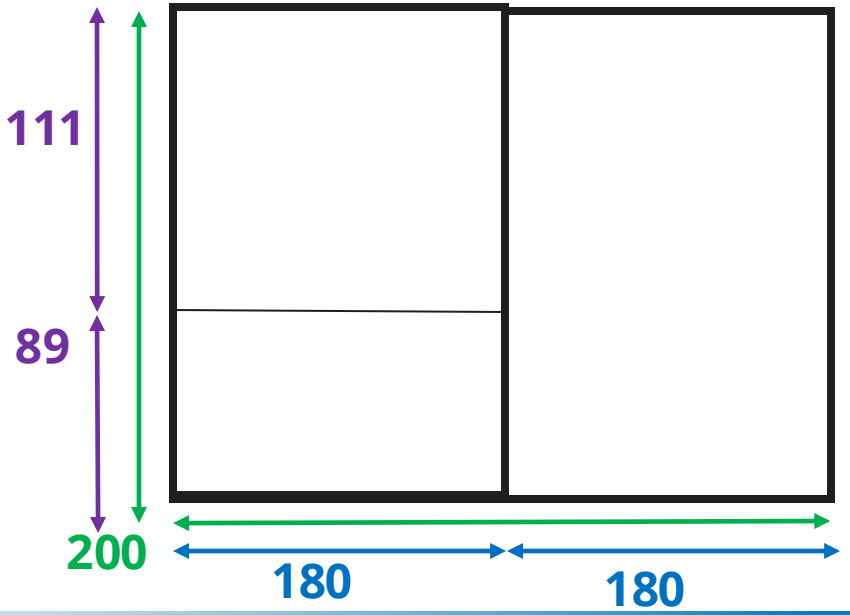




| | | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Department Name | Area ($ft^2$) | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

# Example 6.6



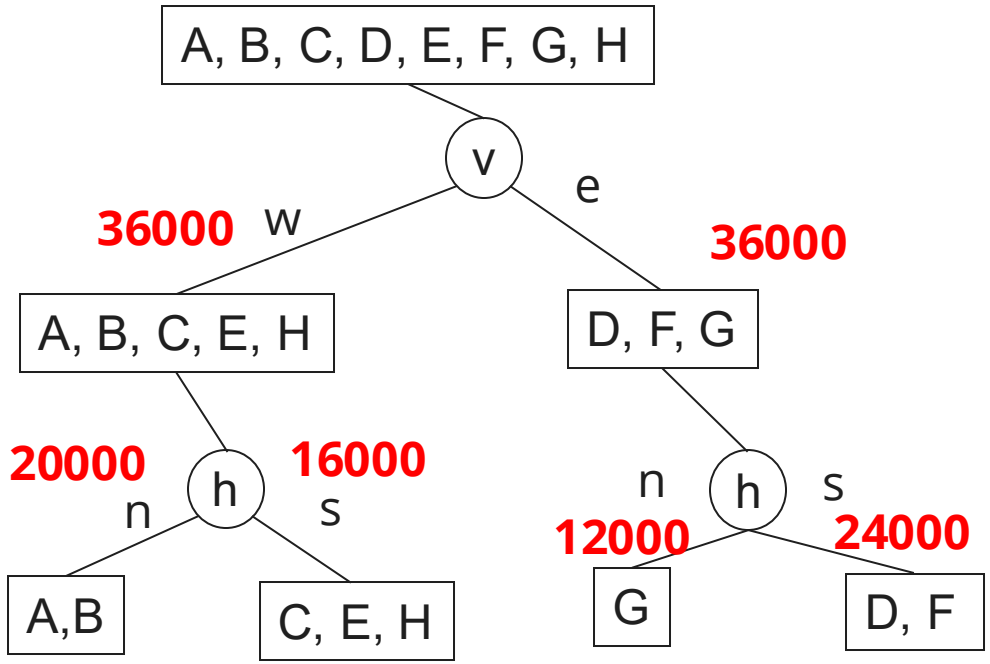

| | | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Department Name | Area ($ft^2$) | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |



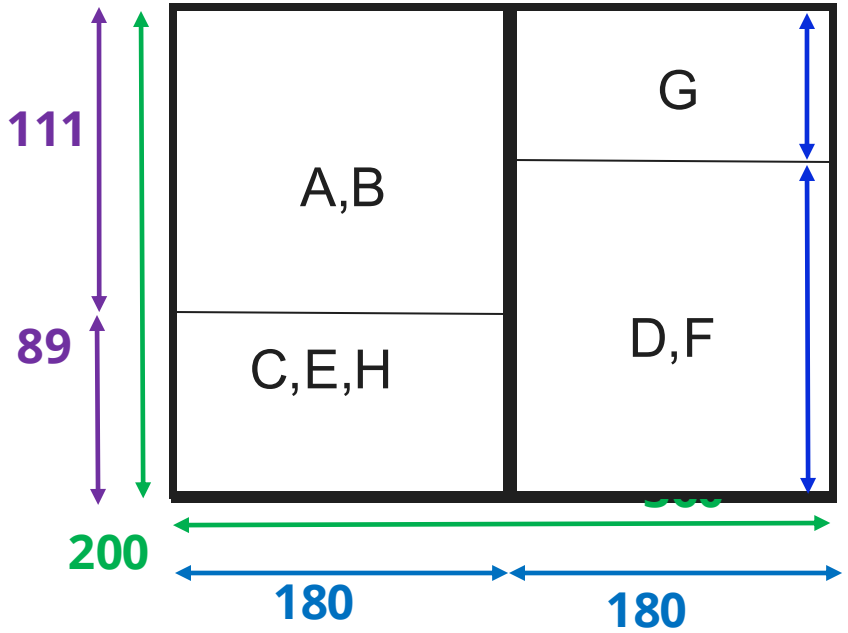



THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

# Example 6.6




| Department Name | Area ($ft^2$) | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

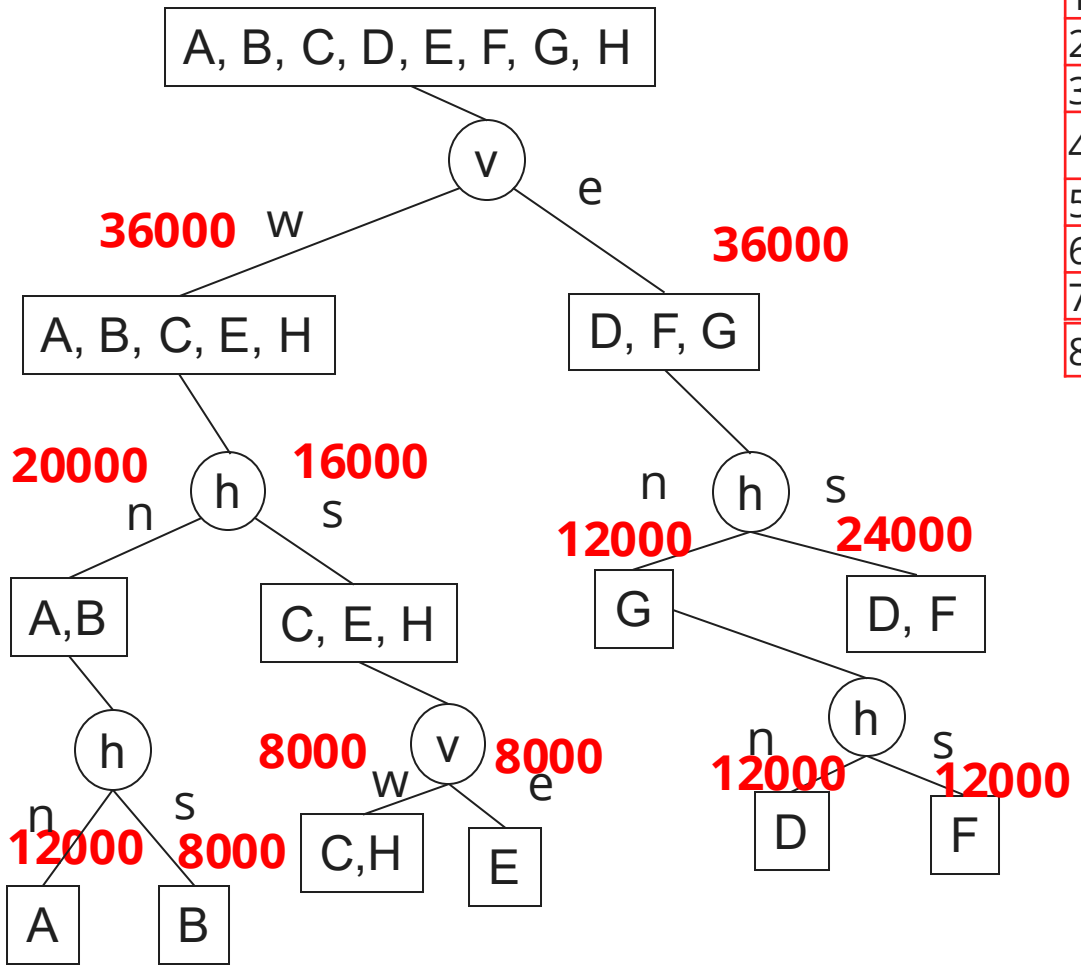


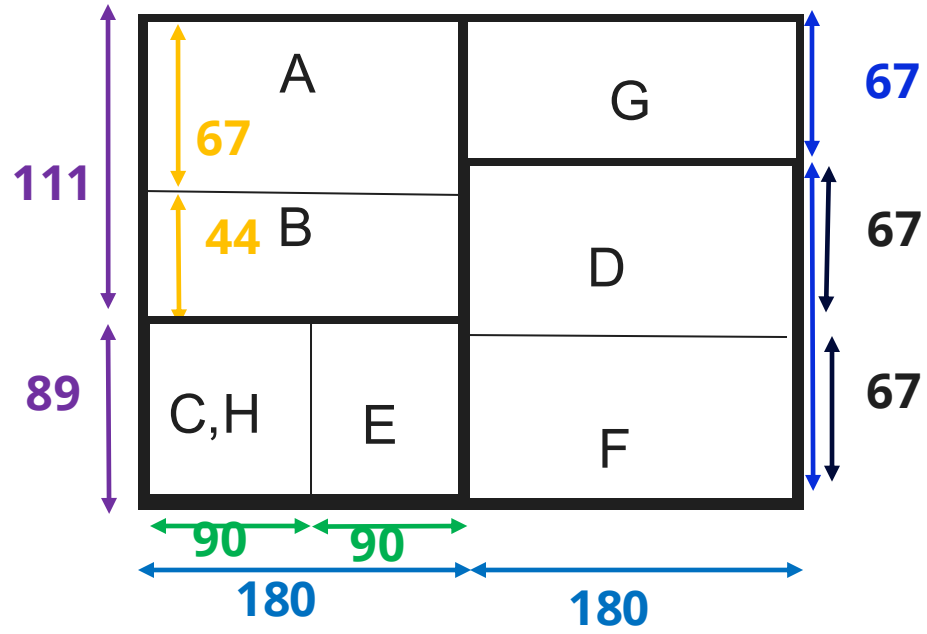



THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

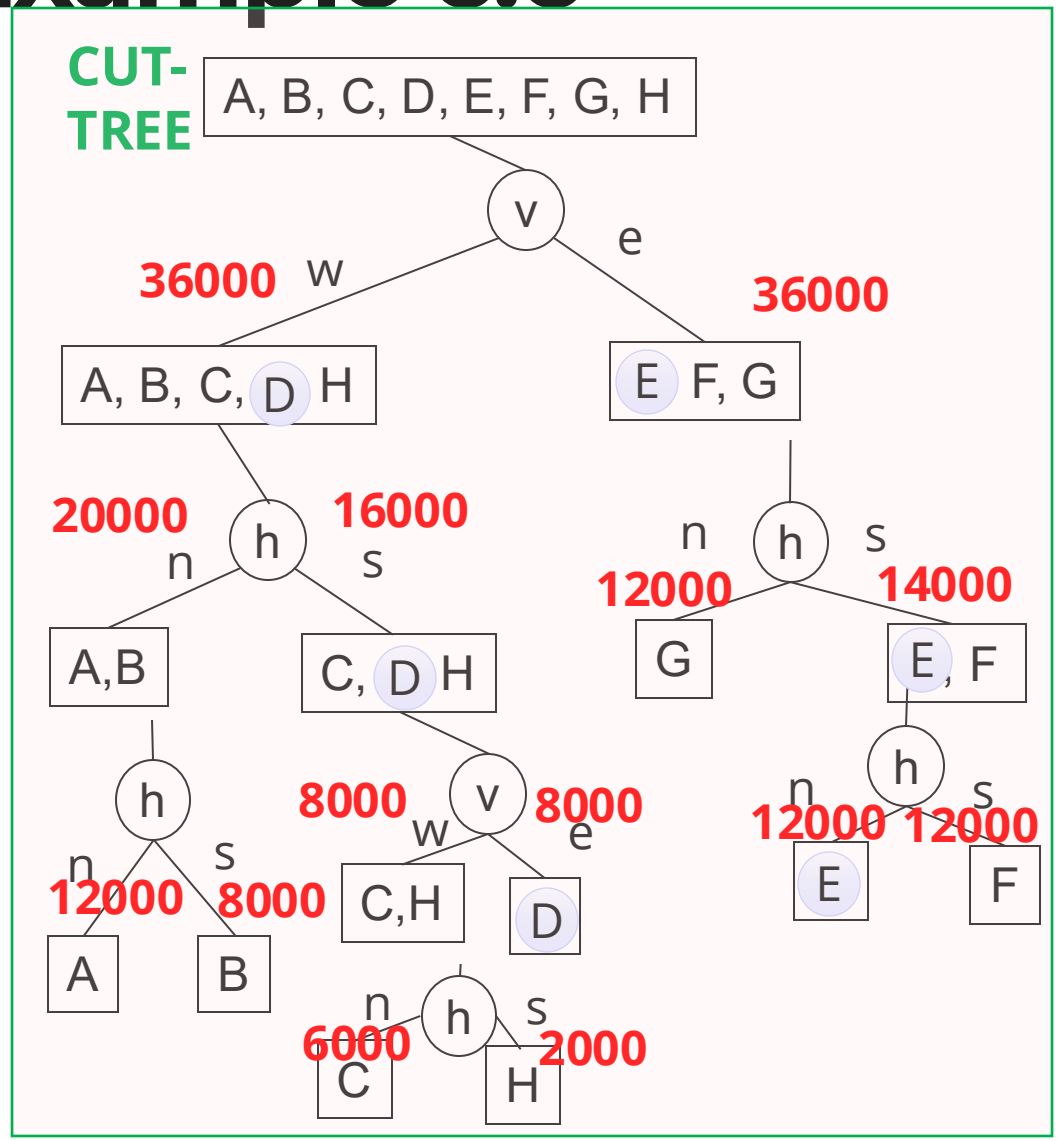
# Example 6.6

| | | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Department Name | Area ($ft^2$) | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |



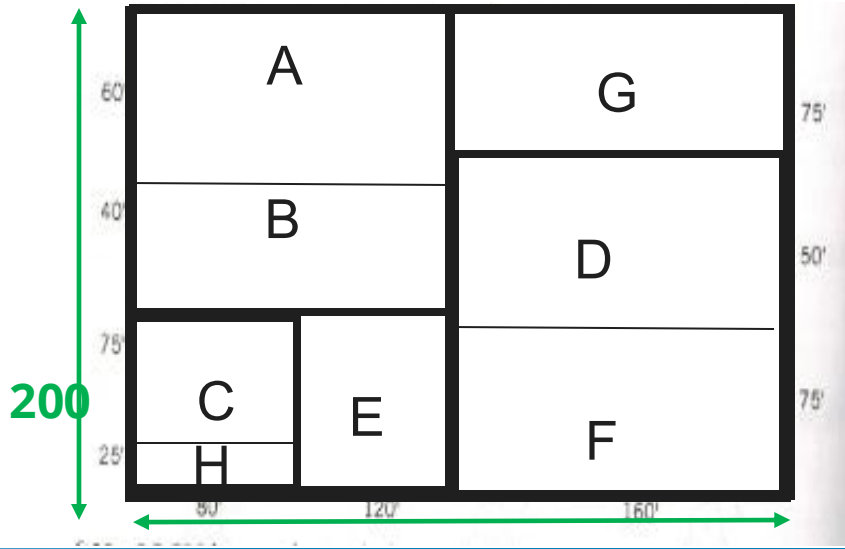



# Example 6.6




| | | FLOW | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Department Name | Area ($ft^2$) | A | B | C | D | E | F | G | H |
| 1. A:Receiving | 12,000 | 0 | 45 | 15 | 25 | 10 | 5 | 0 | 0 |
| 2 B:Milling | 8,000 | 0 | 0 | 0 | 30 | 25 | 15 | 0 | 0 |
| 3 C:Press | 6,000 | 0 | 0 | 0 | 0 | 5 | 10 | 0 | 0 |
| 4 D:Screw m/c. | 12,000 | 0 | 20 | 0 | 0 | 35 | 0 | 0 | 0 |
| 5 E:Assembly | 8,000 | 0 | 0 | 0 | 0 | 0 | 65 | 35 | 0 |
| 6 F:Plating | 12,000 | 0 | 5 | 0 | 0 | 25 | 0 | 65 | 0 |
| 7 G:Shipping | 12,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 H:Dummy | 2,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**What if department exchange D-E is suggested? Replace D-E in cut-tree and recalculate the cuts**

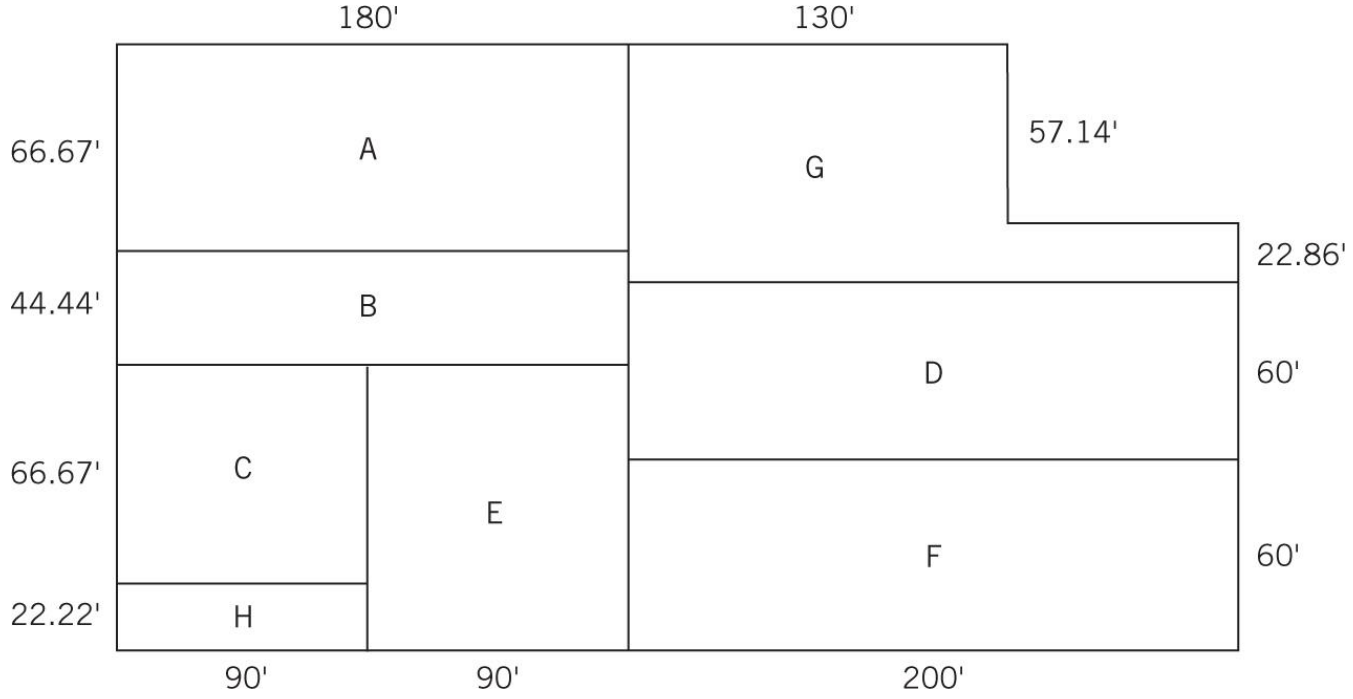





THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

# Example 6.6

Non-rectangular shapes can be handled if they are “reasonable”




Each tree decisions come from search strategy (e.g. simulated annealing)

# Simulated Annealing

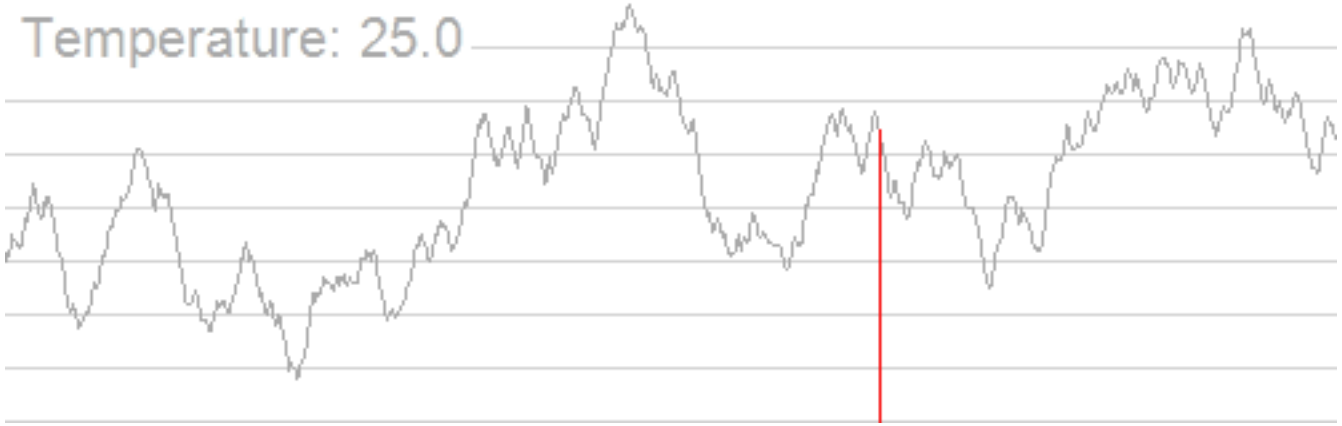
**Simulated annealing** (**SA**) is a probabilistic technique for approximating the global optimum of a given function. Specifically, it is a metaheuristic to approximate global optimization in a large search space. It is often used when the search space is discrete (e.g., all tours that visit a given set of cities). For problems where finding an approximate global optimum is more important than finding a precise local optimum in a fixed amount of time, simulated annealing may be preferable to alternatives such as gradient descent.

- Let $s = s_0$
- For $k = 0$ through $k_{max}$ (exclusive):
  - $T \leftarrow$ temperature( $1 - (k+1)/k_{max}$ )
  - Pick a random neighbour, $s_{new} \leftarrow$ neighbour($s$)
  - If $P(E(s), E(s_{new}), T) \geq$ random(0, 1):
    - $s \leftarrow s_{new}$
- Output: the final state $s$

https://en.wikipedia.org/wiki/Simulated_annealing

# Simulated Annealing

At each step, the simulated annealing heuristic considers some neighbouring state *s′* of the current state *s*, and probabilistically decides between moving the system to state *s′* or staying in state *s*. These probabilities ultimately lead the system to move to states of lower energy. Typically this step is repeated until the system reaches a state that is good enough for the application, or until a given computation budget has been exhausted.




https://en.wikipedia.org/wiki/Simulated_annealing

https://www.youtube.com/watch?v=FyyVbuLZav8

# Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

# MULTIPLE

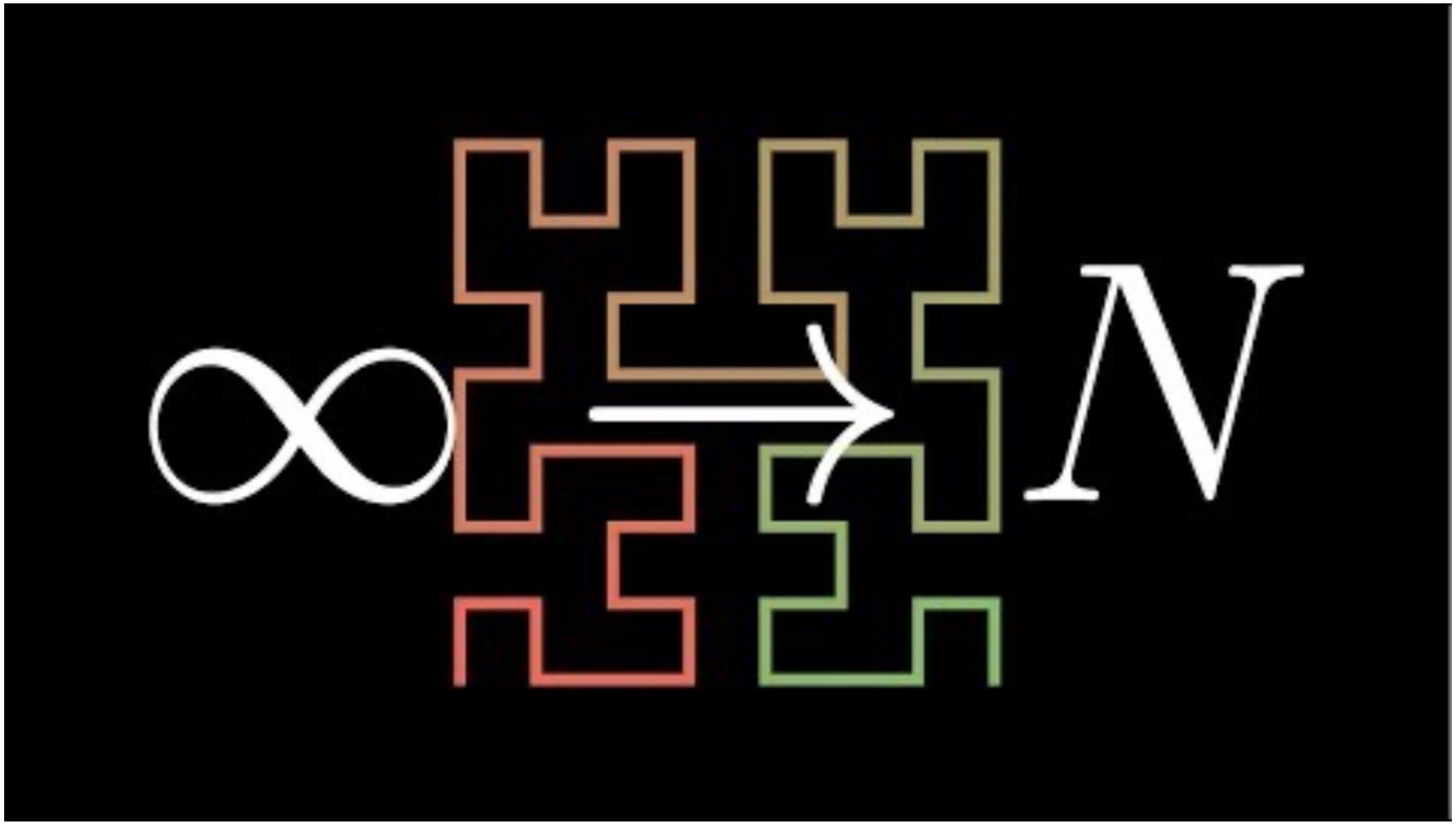
- Layout vector denotes order of department placement
- Fill sequence is determined by space filling curve (SFC)
- Can exchange departments by changing their position in layout vector
- SFC can bypass fixed departments or areas
- Can also be used as a construction procedure

# MULTIPLE (MULTI-floor Plant Layout Evaluation)

- Can also be used for single floors
- Similar to CRAFT
    - Uses from-to chart
    - Objective function identical to CRAFT
    - Not restricted to rectangular shapes
- Discrete representation
- Improvement-type algorithm
- Uses two-way exchanges
- Steepest descent procedure
- Like MicroCRAFT and BLOCPLAN, MULTIPLE can exchange non-adjacent departments
- Uses space-filling curves to place depts.

# MULTIPLE

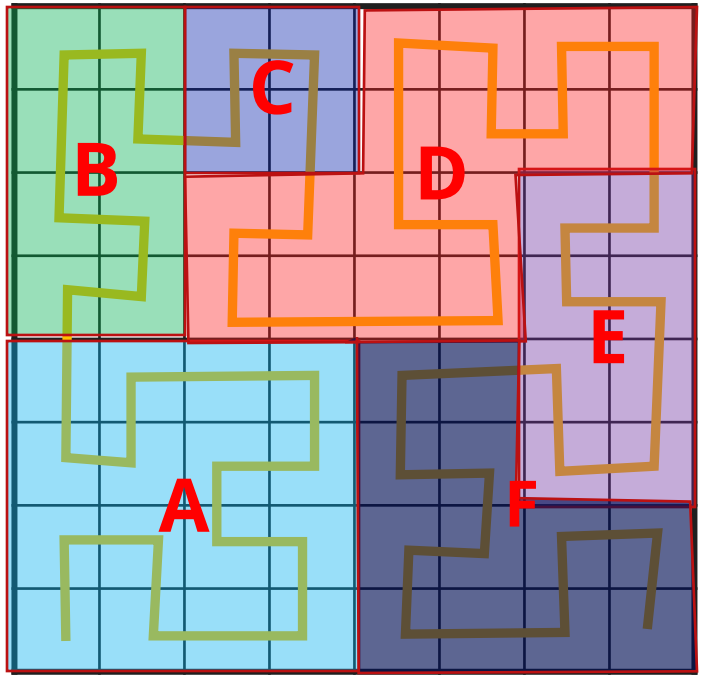
- What's Hilbert Curve?



# Example

- Layout Vector
- A-B-C-D-E-F and their corresponding space requirements are
- Areas from A - F: 16,8,4,16,8,12
- Used to construct heuristic procedure for routing and partition problems and for determining efficient locations for items in storage rack
- Start from grid 1 and assign the first 16 grids (along the SFC) to department A, the next eight grids to department B, and so on.

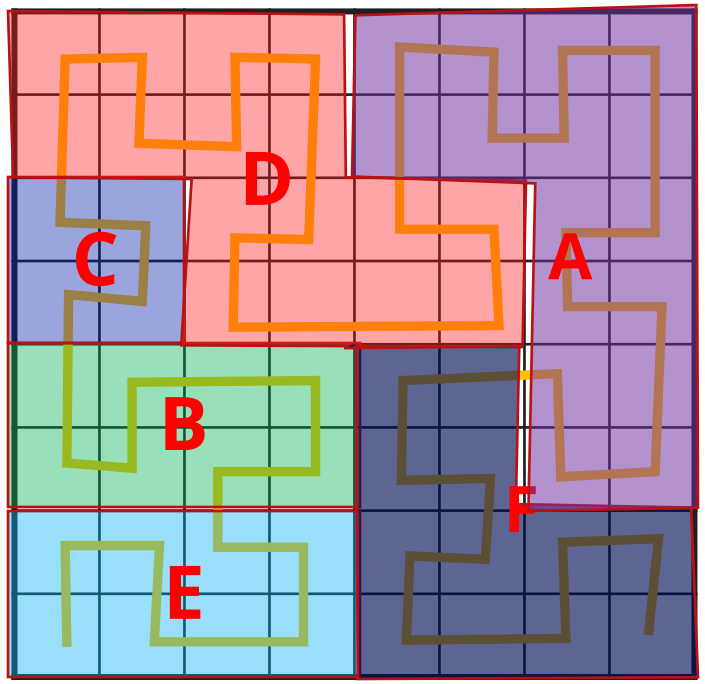
See pp. 338 and references therein for the generation of Hilbert curve.




**Hilbert space filling curve**

# Example

- If department exchange 1-5 is suggested, we update Layout Vector
- **E**-B-C-D-**A**-F and their corresponding space requirements are
- **8**,8,4,16,**16**,12




Hilbert space filling curve

- For departments with fixed locations, skip such grids while following the curve

# Layout Algorithms

| Algorithm | Space requirement | Representation | Techniques |
|---|---|---|---|
| Pairwise exchange method | Cannot handle | | |
| Graph-based method | Cannot handle | | Planar graph |
| CRAFT | Can handle | Discrete | grids |
| MCRAFT | Can handle | Continuous | bands |
| BLOCPLAN | Can handle | Continuous | bands |
| MIP | Can handle | Continuous | Math programming |
| LOGIC | Can handle | Continuous | Horizontal/vertical cuts |
| MULTIPLE | Can handle | Discrete | Space filling curve |

# Layout Algorithms

| Algorithm | Objective Function | Type | Restrictions of exchange |
|---|---|---|---|
| Pairwise exchange method | Distance-based | Improvement | Adjacent departments |
| Graph-based method | Adjacency-based | Construction | |
| CRAFT | Distance-based | Improvement | Adjacent departments |
| MCRAFT | Distance-based | Improvement | No restriction |
| BLOCPLAN | Both | Improvement | No restriction |
| MIP | Distance-based | Construction | |
| LOGIC | Distance-based | Improvement | No restriction |
| MULTIPLE | Distance-based | Improvement | No restriction |

# Multifloor Facility Layout

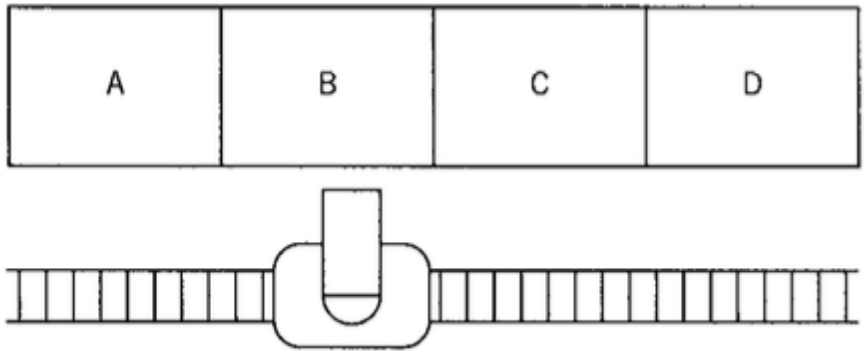
- Most industrial facilities are single-floor facilities, such as manufacturing, warehousing, etc.
- In certain situation, multifloor is needed due to resource constraints.
- A more complex planning problem, mainly due to the vertical travel between floors.
- More complex interactions, and constraints.
- We will not go into details for multifloor layout. See Section 6.7 for a discussion.

# “Principles” for Layout Alternatives

- Do not favor one solution or reject a solution.
- Set a time limit and use the available time efficiently.
- Don’t “fail to see the forest for the trees”.
- Think big, then think little.
- Refer to analogous problems.
- Involve operating people, management, experienced people, inexperienced people, those who oppose change, and those who promote change.
- Think long range.

# Examples:

Four equal-sized machines are served by an automated guided vehicle (AGV) on a linear bidirectional track, as shown in the figure below. Each machine block is 30' * 30'. The product routine information and required production rate are given in the table below. Determine a layout arrangement based on the pairwise exchange method. Assume that the pickup/delivery stations are located at the midpoint of the machine edge along the AGV track.




| Product | Processing Sequences | Weekly Production |
|---|---|---|
| 1 | BDCAC | 300 units |
| 2 | BDAC | 700 units |
| 3 | DBDCAC | 900 units |
| 4 | ABCA | 200 units |

Four equal-sized machines are served by an automated guided vehicle (AGV) on a linear bidirectional track, as shown in the figure below. Each machine block is 30' * 30'. The product routine information and required production rate are given in the table below. Determine a layout arrangement based on the pairwise exchange method. Assume that the pickup/delivery stations are located at the midpoint of the machine edge along the AGV track.

Flow-between chart

| Machine | A | B | C | D |
|---|---|---|---|---|
| A | – | | | |
| B | | – | | |
| C | | | – | |
| D | | | | – |

| Product | Processing Sequences | Weekly Production |
|---|---|---|
| 1 | BDCAC | 300 units |
| 2 | BDAC | 700 units |
| 3 | DBDCAC | 900 units |
| 4 | ABCA | 200 units |

Arrangements/Costs:

A B C D: 30(200 + 200 + 1200) + 60(3300 + 2800) + 90(700) = 477,000

B A C D:

C B A D:

D B C A:

Switch A and D:

B D C A: 30(2800 + 1200 + 3300) + 60(200 + 700) + 90(200) = 291,000

C B D A: 30(200 + 2800 + 700) + 60(1200 + 200) + 90(3300) = 492,000

A B C D: 30(200 + 200 + 1200) + 60(3300 + 2800) + 90(700) = 477,000

Switch B and D:

C D B A: 30(1200 + 2800 + 200) + 60(200 + 700) + 90(3300) = 477,000

A D C B: 30(700 + 2800 + 200) + 60(3300 + 2800) + 90(200) = 495,000

# Definition of Material Handling

- Material handling accounts for 25% of all employees, 55% of all factory space, 87% of the production time and between 15% and 70% of the total cost of a manufactured product.
- Many improvements can be achieved in Material Handling, resulting in cost savings.
- Ideal Goal: “totally eliminate” material handling activities
- Practical Goal: reduce the amount of handling

# Definition of Material Handling

- Definition 1: Material handling is the art and science of moving, storing, protecting and controlling materials throughout the process of their manufacture, distribution, consumption, and disposal.

- Involves defining the problem, collecting and analyzing data, generating alternative solutions, evaluating alternatives, selecting and implementing preferred alternative(s), performing periodic reviews

- Material Handling systems cannot be explicitly designed based solely on scientific formulas or mathematical models. Knowledge of what is “right and wrong” based on significant practical experience in the field.

# Definition of Material Handling

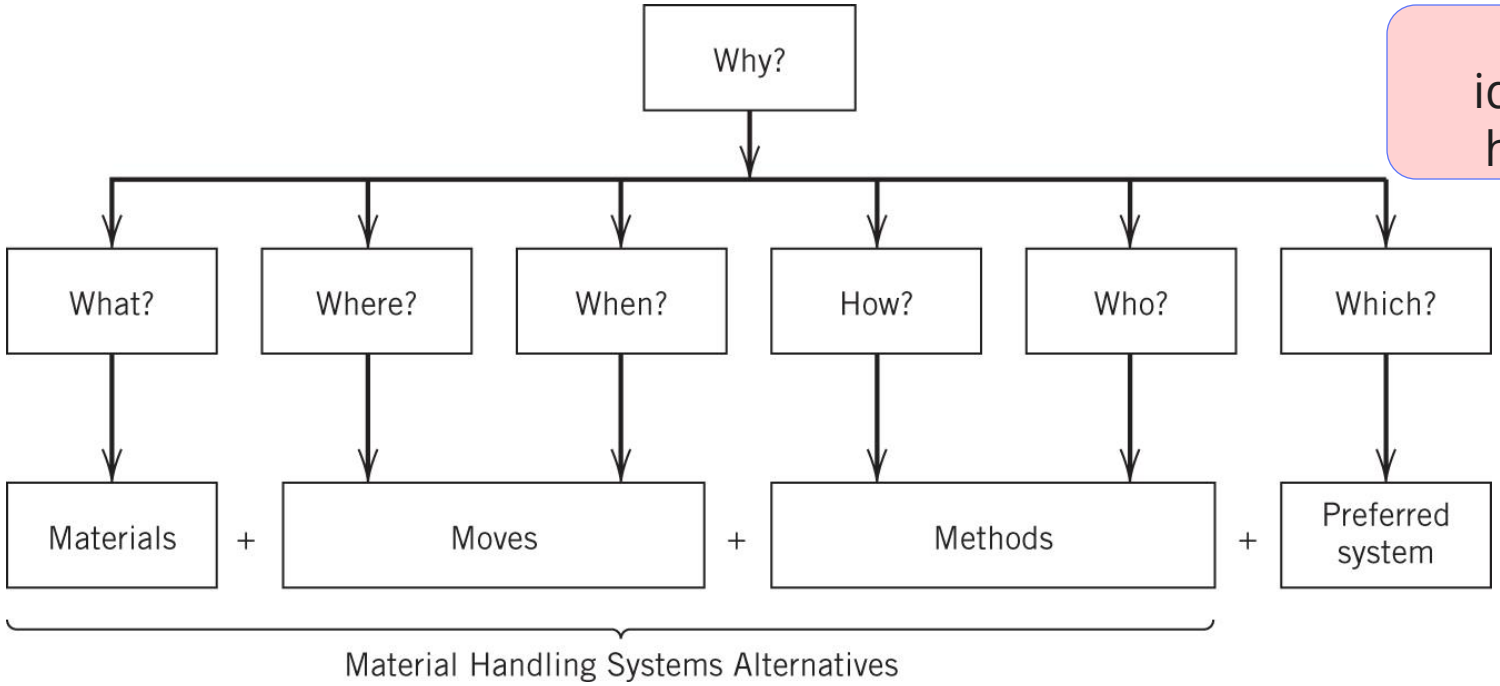
Definition 2: Material handling means providing:

- the right amount — **how much inventory is needed.**
- of the right material, — **Accurate identification: RFID, bar code**
- in the right condition, — **In which state customer desires to receive material (packed, unpacked, sorted, painted, unpainted…)**
- at the right sequence, — **Activities of manufacturing or distribution operation**
- at the right orientation, — **Positioning material for ease of handling**
- in the right place, — **Transportation and storage: directly to point of use**
- in the right time, — **On-time delivery, neither early nor late**
- and for the right cost, — **Efficient MH systems at reasonable cost**
- by using the right methods. — **Requirement-driven vs. solution-driven**

# Definition of Material Handling

Developing alternative materials handling systems by way of procedures and algorithms is **much more difficult than for layout**

This is due to the almost infinite possibilities of items to move and the vast numbers of alternative material handling equipment and methods




Framework for identifying material handling solutions

# Definition of Material Handling

1. Planning
2. Standardization
3. Work
4. Ergonomic
5. Unit Load
6. Space Utilization
7. System
8. Automation
9. Environmental
10. Life Cycle Cost

**Often no mathematical model can provide a solution to Material Handling problems. These principles provide statements of fundamentals of material handling practice.**

Material Handling Institute

http://www.mhi.org/

## 1 Planning Principle

All material handling should be the result of a deliberate plan where the needs, performance objectives and functional specification of the proposed methods are completely defined at the outset.

**Definition:** *A plan is a prescribed course of action that is defined in advance of implementation. In its simplest form a material handing plan defines the material (what) and the moves (when and where); together they define the method (how and who).*

### Key Points:

- The plan should be developed in consultation between theplanner(s) and all who will use and benefit from the equipment to be employed.
- Success in planning large scale material handling projects generally requires a team approach involving suppliers, consultants when appropriate, and end user specialists from management, engineering, computer and

out sacrificing needed flexibility , modularity and throughput.anticipation of changing future requirements

**Definition:** *Standardization means less variety and customization in the methods and equipment employed.*

### Key Points:

- The planner should select methods and equipment that can perform a variety of tasks under a variety of operating conditionsand in
- Standardization applies to sizes of containers and other load forming components as well as operating procedures and equipment.
- Standardization, flexibility and modularity must not be incompatible.

## 3 Work Principle

Material handling work should be minimized without sacrificing productivity or the level of service required of the operation.



THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

# Planning Principle

- Definition: A plan is a prescribed course of action that is defined in advance of implementation. **In its simplest form a material handing plan defines the material (what) and the moves (when and where); together they define the method (how and who).**

# Planning Principle

- Key Points
  - The plan should be developed in consultation between the planner(s) and all who will use and benefit from the equipment to be employed.
  - Success in planning large scale material handling projects generally requires **a team approach** involving suppliers, consultants when appropriate, and end user specialists from management, engineering, computer and information systems, finance and operations.
  - The material handling plan should reflect the strategic objectives of the organization as well as the more immediate needs.

# Planning Principle

- Key Points
  - The plan should document existing methods and problems, physical and economic constraints, and future requirements and goals.
  - The plan should promote concurrent engineering of product, process design, process layout, and material handling methods, as opposed to independent and sequential design practices.

# Planning Principle: Lost at sea example

## Situation

You are adrift on a private yacht in the South Pacific. As a consequence of a fire of unknown origin, much of the yacht and its contents have been destroyed. The yacht is now slowly sinking. Your location is unclear because of the destruction of critical navigational equipment and because you and the crew were distracted trying to bring the fire under control. Your best estimate is that you are approximately one thousand miles south-southwest of the nearest land.

Below is a list of fifteen items that are intact and undamaged after the fire. In addition to these articles, you have a serviceable, rubber life raft with oars large enough to carry yourself, the crew, and all the items listed below. The total contents of all survivors' pockets are a package ofcigarettes, several books of matches, and five one-dollar bills.Your task is to rank the fifteen items below in terms of their importance to your survival. Place the number 1 by the most important item, the number 2 by the second most important, and soon through number 15, the least important. You will do this twice: once alone (10 min.) and once as a group (20 min.).

https://newmfgalliance.org/wp-content/uploads/2020/03/lost-at-sea-instructions-8-19-2015.pdf?

# Planning Principle: Lost at Sea example

| Items | Step 1<br>Your Individual Ranking | Step 2<br>Your Team Ranking | Step 3<br>Coast Guard Ranking | Step 4<br>Difference between Step 1 & 3 | Step 5<br>Difference between Step 2 & 3 |
|---|---|---|---|---|---|
| Sextant | | | | | |
| Shaving Mirror | | | | | |
| Mosquito Netting | | | | | |
| 25 liter container of Water | | | | | |
| Case of Army Rations | | | | | |
| Maps of the Pacific Ocean | | | | | |
| Floating Seat Cushion | | | | | |
| 10 liter can of Oil/Petrol mixture | | | | | |
| Small Transistor Radio | | | | | |
| 20 sq. ft. of opaque Plastic Sheeting | | | | | |
| Can of Shark Repellent | | | | | |
| One bottle rubbing alcohol | | | | | |
| 15 ft. of Nylon Rope | | | | | |
| 2 boxes of Chocolate Bars | | | | | |
| An ocean Fishing Kit & Pole | | | | | |
| | | | **TOTALS** | Your Score | Team Score |

# Planning Example

| Item | Coast Guard Ranking | Reasoning |
|---|---|---|
| Sextant | 15 | Useless without the relevant tables & a chronometer. |
| Shaving Mirror | 1 | Of all the items, the mirror is absolutely critical. It is the most powerful tool you have for communicating your presence. In sunlight, a simple mirror can generate five to seven million candlepower of light. The reflected sunbeam can even be |

## Scores

| | | |
|---|---|---|
| 00 - 25 | Excellent | You demonstrated great survival skills. Rescued! |
| 26 - 32 | Good | Above average results. Good survival skills. Rescued! |
| 33 - 45 | Average | Seasick, hungry and tired. Rescued! |
| 46 - 55 | Fair | Dehydrated and barely alive. It was tough, but rescued! |
| 56 - 70 | Poor | Rescued, but only just in time! |
| 71+ | Very Poor | Oh dear, your empty raft is washed up on a beach, weeks after the search was called off. |

| | | |
|---|---|---|
| 15 ft. of Nylon Rope | 8 | prevent being washed overboard. There are a variety of other uses but non high on the list for survival. |
| 2 boxes of Chocolate Bars | 6 | Your reserve food supply. |
| An ocean Fishing Kit & Pole | 7 | Ranked lower than chocolate as there is no guarantee you'll catch fish. Pole could be used as a tent pole. |

# Standardization Principle

- Definition: Standardization means less variety and customization in the methods and equipment employed.
- Key Points
  - **The planner should select methods and equipment that can perform a variety of tasks under a variety of operating conditions and in anticipation of changing future requirements.**
  - Standardization applies to sizes of containers and other load forming components as well as operating procedures and equipment.
  - Standardization, flexibility and modularity must be compatible

# Work Principle

- Definition: The measure of work is material handling flow (volume, weight or count per unit of time) multiplied by the distance moved.
- Key Points
  - Material handling work should be *minimized without sacrificing productivity or the level of service required of the operation.*
  - Simplifying processes by reducing, combining, shortening or eliminating unnecessary moves will reduce work.
  - Consider each pickup and set-down, or placing material in and out of storage, as distinct moves and components of the distance moved.

# Work Principle

- Key Points
  - Process methods, operation sequences and process/equipment layouts should be prepared that support the work minimization objective.
  - Where possible, gravity should be used to move materials or to assist in their movement while respecting consideration of safety and the potential for product damage.
  - The shortest distance between two points is a straight line.

# Ergonomic Principle

- Definition:  Ergonomics is the science that seeks to adapt work or working conditions to suit the abilities of the worker.
- Key Points
  - Human capabilities and limitations must be recognized and respected in the design of material handling tasks and equipment to ensure safe and effective operations.
  - Equipment should be selected that eliminates repetitive manual labor and which effectively interacts with human operators and users.
  - The ergonomic principle embraces both physical and mental tasks.
  - The material handling workplace and the equipment employed to assist in that work must be designed so they are safe for people.

# Unit Load Principle

- Definition: A unit load is one that can be stored or moved as a single entity at one time, such as a pallet, container or tote, regardless of the number of individual items that make up the load.
- Key Points
  - Unit loads shall be appropriately sized and configured in a way which achieves the material flow and inventory objectives at each stage in the supply chain.
  - Less effort and work is required to collect and move many individual items as a single load than to move many items one at a time.
  - Load size and composition may change as material and product moves through stages of manufacturing and the resulting distribution channels.
  - Large unit loads are common both pre and post manufacturing in the form of raw materials and finished goods.

# Unit Load Principle

- Key Points
  - During manufacturing, smaller unit loads, including as few as one item, yield less in-process inventory and shorter item throughput times.
  - Smaller unit loads are consistent with manufacturing strategies that embrace operating objectives such as flexibility, continuous flow and just-in-time delivery.
  - Unit loads composed of a mix of different items are consistent with just-in-time and/or customized supply strategies so long as item selectivity is not compromised.

# Space Utilization

- Definition: Space in material handling is three dimensional and therefore is counted as cubic space.
- Key Points
  - Effective and efficient use must be made of all available space.
  - In work areas, cluttered and unorganized spaces and blocked aisles should be eliminated.
  - In storage areas, the objective of maximizing storage density must be balanced against accessibility and selectivity.
  - When transporting loads within a facility, the use of overhead space should be considered as an option.

# System Principle

- Definition:  A system is a collection of interacting and/or interdependent entities that form a unified whole.
- Key Points
  - Material movement and storage activities should be fully integrated to form a coordinated, operational system which spans receiving, inspection, storage, production, assembly, packaging, unitizing, order selection, shipping, transportation and the handling of returns.
  - Systems integration should encompass the entire supply chain including reverse logistics. It should include suppliers, manufacturers, distributors and customers.
  - Inventory levels should be minimized at all stages of production and distribution while respecting considerations of process variability and customer service.

# System Principle

- Key Points
  - Information flow and physical material flow should be integrated and treated as concurrent activities.
  - Methods should be provided for easily identifying materials and products, for determining their location and status within facilities and within the supply chain and for controlling their movement.
  - Customer requirements and expectations regarding quantity, quality, and on-time delivery should be met without exception.

# Automation Principle

- Definition: Automation is a technology concerned with the application of electro-mechanical devices, electronics and computer-based systems to operate and control production and service activities. It suggests the **linking of multiple mechanical operations** to create a system that can be controlled by programmed instructions.

# Automation Principle

- Key Points
  - Material handling operations should be mechanized and/or automated where feasible to improve operational efficiency, increase responsiveness, improve consistency and predictability, decrease operating costs and to eliminate repetitive or potentially unsafe manual labor.
  - Pre-existing processes and methods should be simplified and/or re-engineered before any efforts at installing mechanized or automated systems.
  - Computerized material handling systems should be considered where appropriate for effective integration of material flow and information management.

# Environmental Principle

- Definition:  Environmental consciousness stems from a desire not to waste natural resources and to **predict and eliminate the possible negative effects** of our daily actions on the environment.

# Environmental Principle

- Key Points
  - Environmental impact and energy consumption should be considered as criteria when designing or selecting alternative equipment and material handling systems.
  - Containers, pallets and other products used to form and protect unit loads should be designed for reusability when possible and/or biodegradability as appropriate.
  - Systems design should accommodate the handling of dunnage, empty containers and other by-products of material handling.
  - Materials specified as hazardous have special needs with regard to spill protection, combustibility and other risks.

# Life Cycle Cost Principle

- Definition: Life cycle costs include all cash flows that will occur between the time the first dollar is spent to plan or procure a new piece of equipment, or to put in place a new method, until that method and/or equipment is totally replaced.

# Life Cycle Cost Principle

- Key Points
  - A thorough economic analysis should account for the entire life cycle of all material handling equipment and resulting systems.
  - Life cycle costs include capital investment, installation, setup and equipment programming, training, system testing and acceptance, operating (labor, utilities, etc.), maintenance and repair, reuse value, and ultimate disposal.
  - A plan for preventive and predictive maintenance should be prepared for the equipment, and the estimated cost of maintenance and spare parts should be included in the economic analysis.

# Life Cycle Cost Principle

- Key Points
  - A long-range plan for replacement of the equipment when it becomes obsolete should be prepared.
  - Although measurable cost is a primary factor, it is certainly not the only factor in selecting among alternatives. Other factors of a strategic nature to the organization and which form the basis for competition in the market place should be considered and quantified whenever possible.

# 8 Wastes of Lean Manufacturing



## DEFECTS

Waste from a product or service failure to meet customer expectations



## OVERPRODUCTION

Waste from making more product than customers demand



## WAITING

Waste from time spent waiting for the next process step to occur



## UNUSED TALENT

Wastes due to underutilization of people's talents, skills, and knowledge



## TRANSPORTATION

Wasted time, resources, and costs when unnecessarily moving products and materials



## INVENTORY

Wastes resulting from excess products and materials that aren't processed



## MOTION

Wasted time and effort related to unnecessary movements by people



## EXTRA-PROCESSING

Wastes related to more work or higher quality than is required

# Unit Load

- **A central component** of any materials handling system is the definition of a unit load

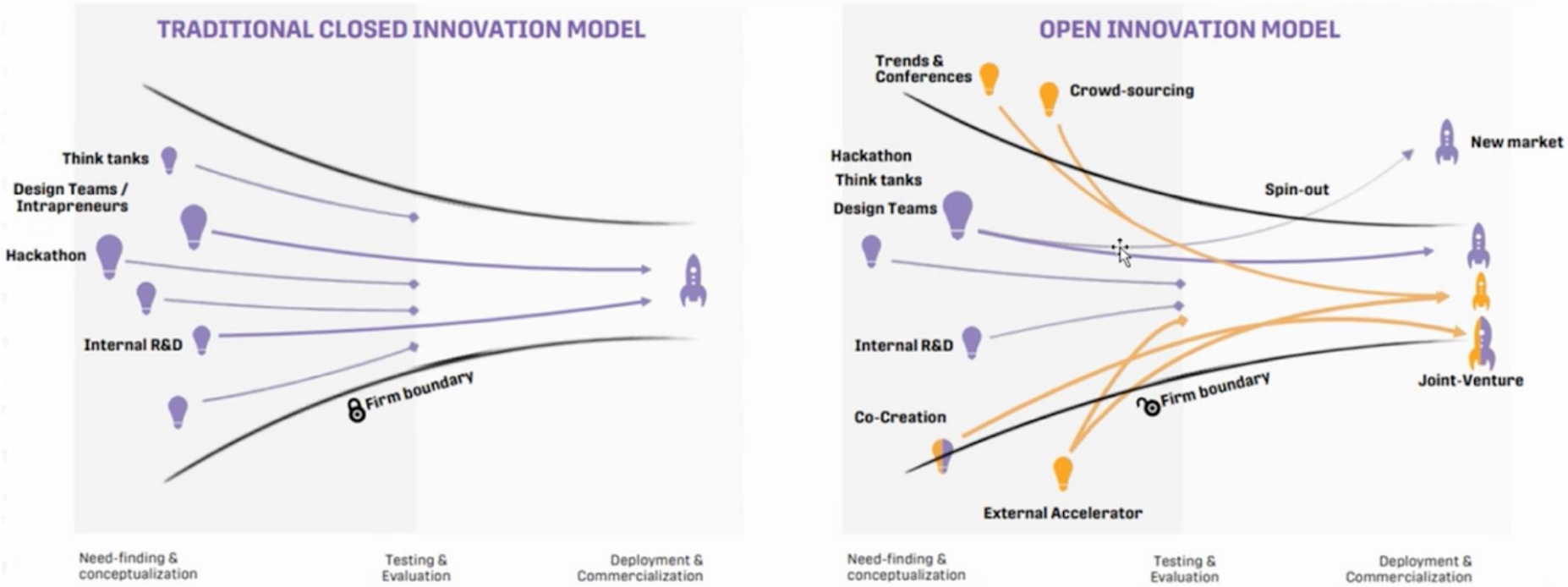
- A unit load is the “picked up and moved between two locations as a single mass” -- A unit load varies





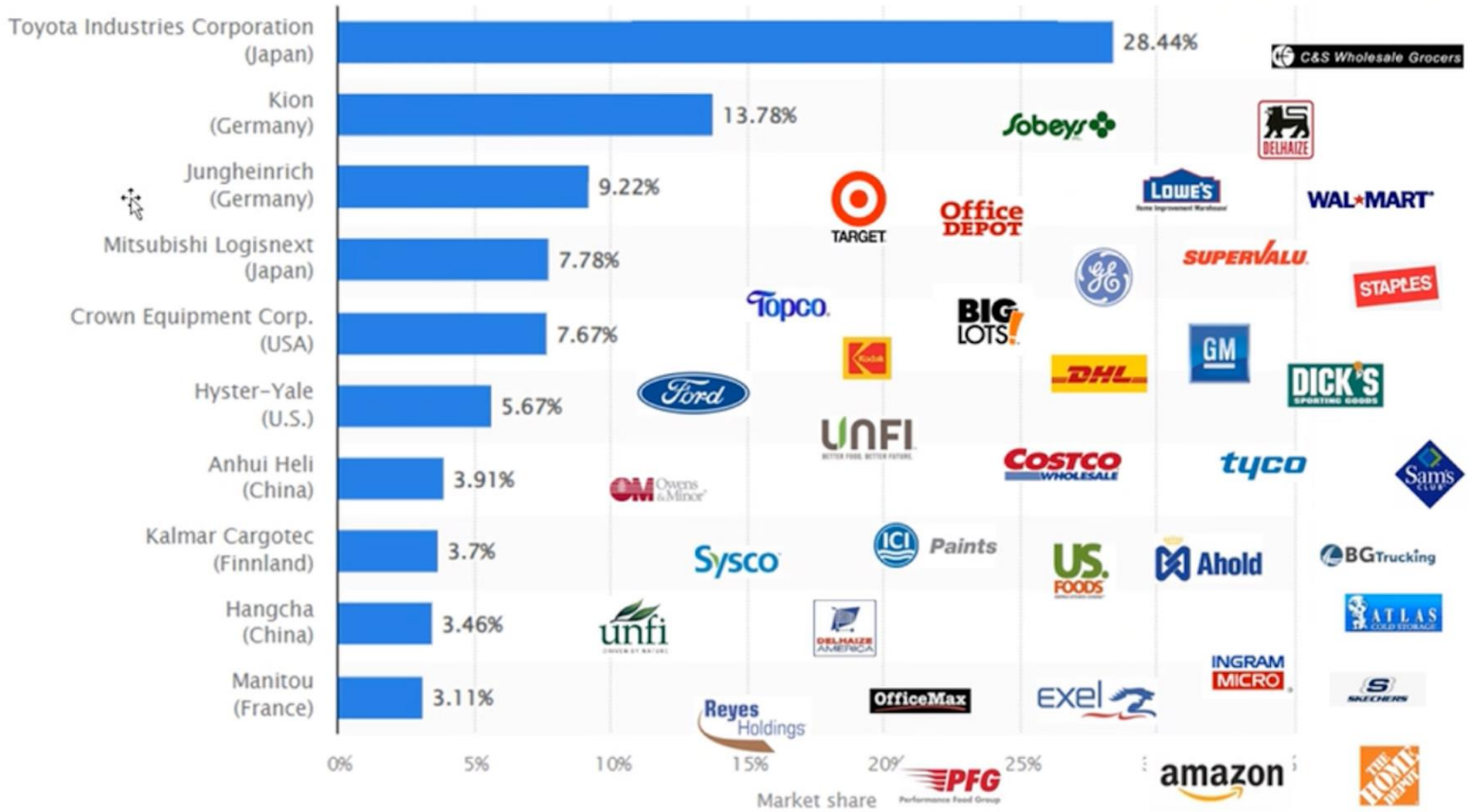


# Unit Load



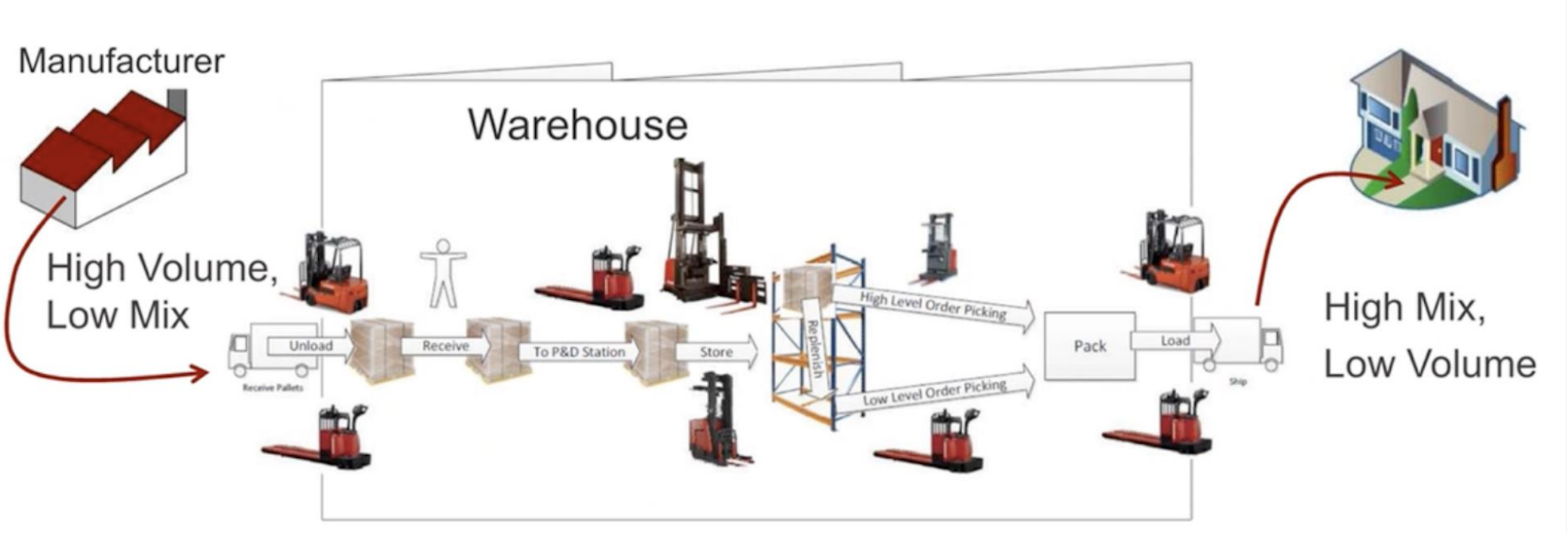

Innovation process and cross pollen

# Unit Load



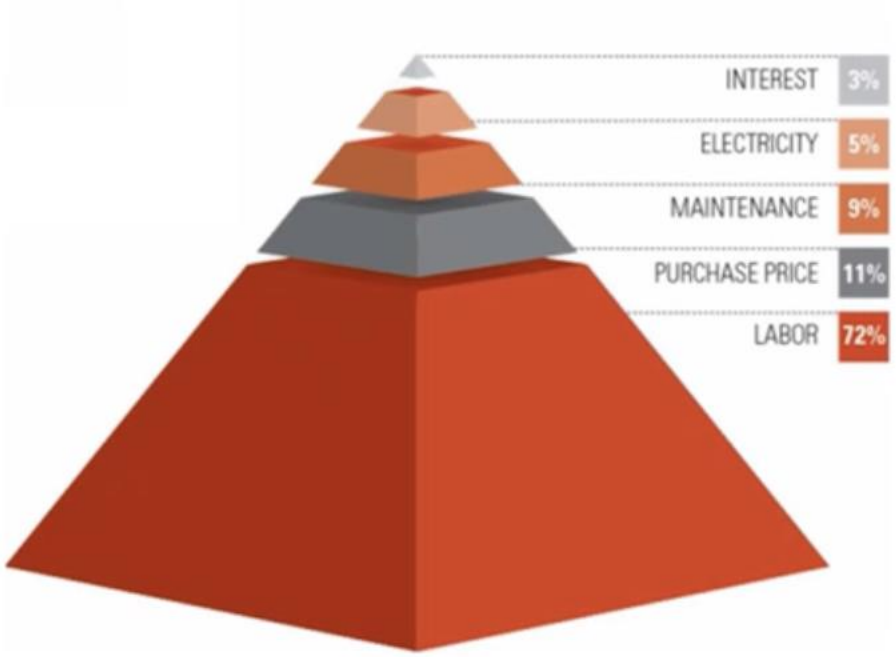

Market share

# Unit Load




- The value the warehouse provides is the service of processing bulk goods into picked orders of only the items the customer needs
- From warehouse perspective, racks are value add and aisles are non-value add

# Unit Load




Typical forklift operating cost

**Customers are demanding solutions that:**

- Improve productivity
- Increase operational efficiency
- Reduce costs
- Maximize asset utilization
- Create safe work environment

Nearly 75% of all warehouse operating expenses are due to labor

# Unit Load



**Challenges to Automate warehouse:**

- High Implementation Cost*
- Scalability*
- Reliability and Maintenance
- System Integration*
- Lack of Expertise and Training
- Lack of Flexibility*
- High Variability in Product Form Factor

* Potential project focus

# Unit Load




**Challenges for customers:**

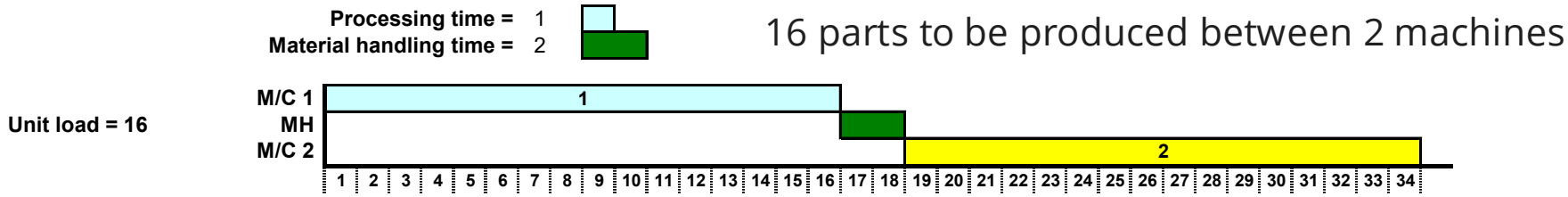
- Price increasing from inflation - 53%
- Talent shortage - 52%
- Customer demands - 51%
- Insight into customer behavior - 50%
- Supply chain disruption/shortage - 50%

# Unit Load

- Small Unit Loads
  - Increase transportation requirements
  - Potentially reduce work-in-process inventory
  - Push carts
- Large Unit Loads
  - Fewer moves
  - Increase work-in-process inventory (items have to accumulate to full unit load size)
  - Require bigger and heavier equipment, wider aisles, and higher floor load capacities

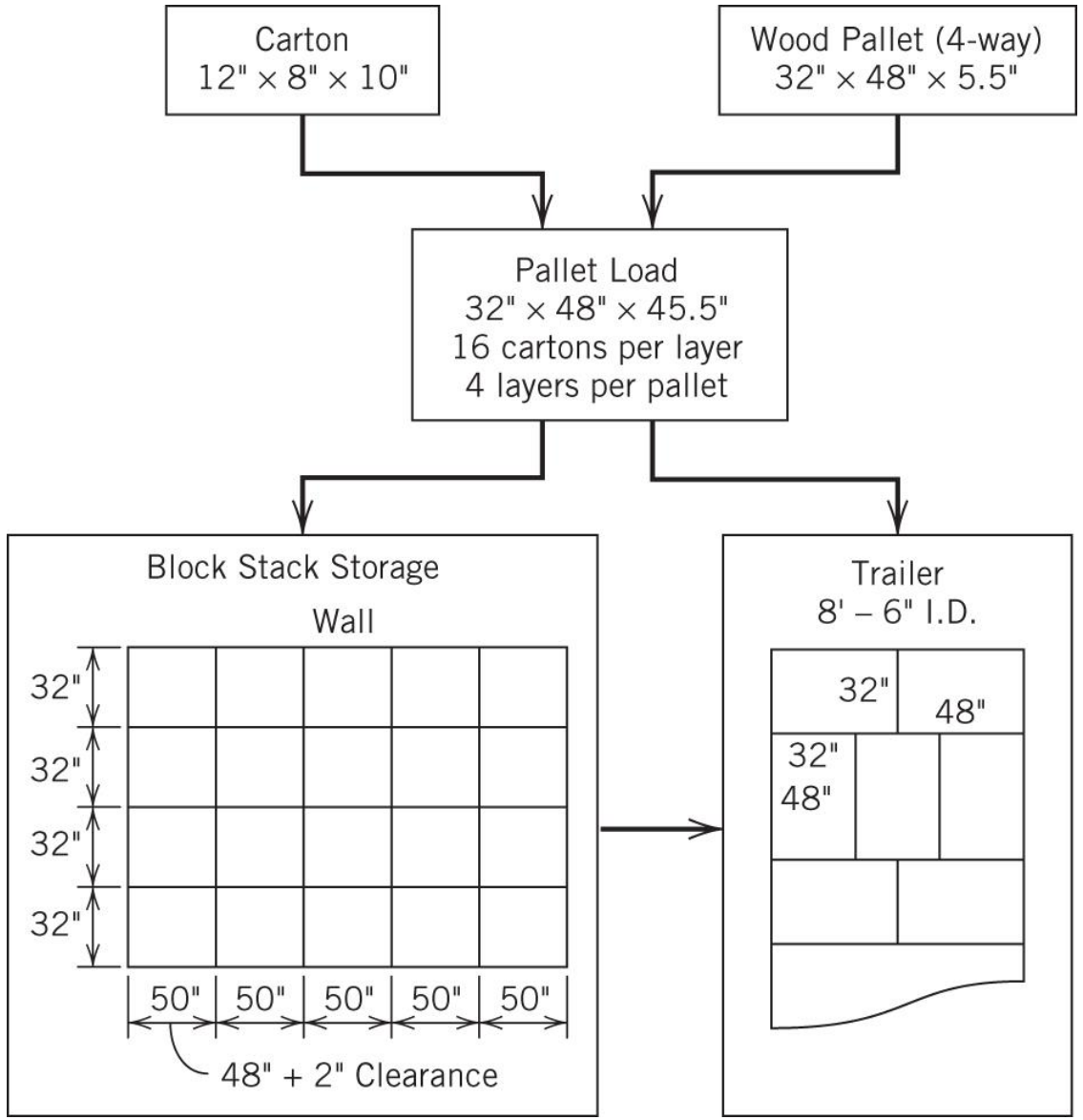
The unit load size specification has a major impact on the specification and operation of the material handling system.

# Unit Load




Effect of unit load size on job completion time.

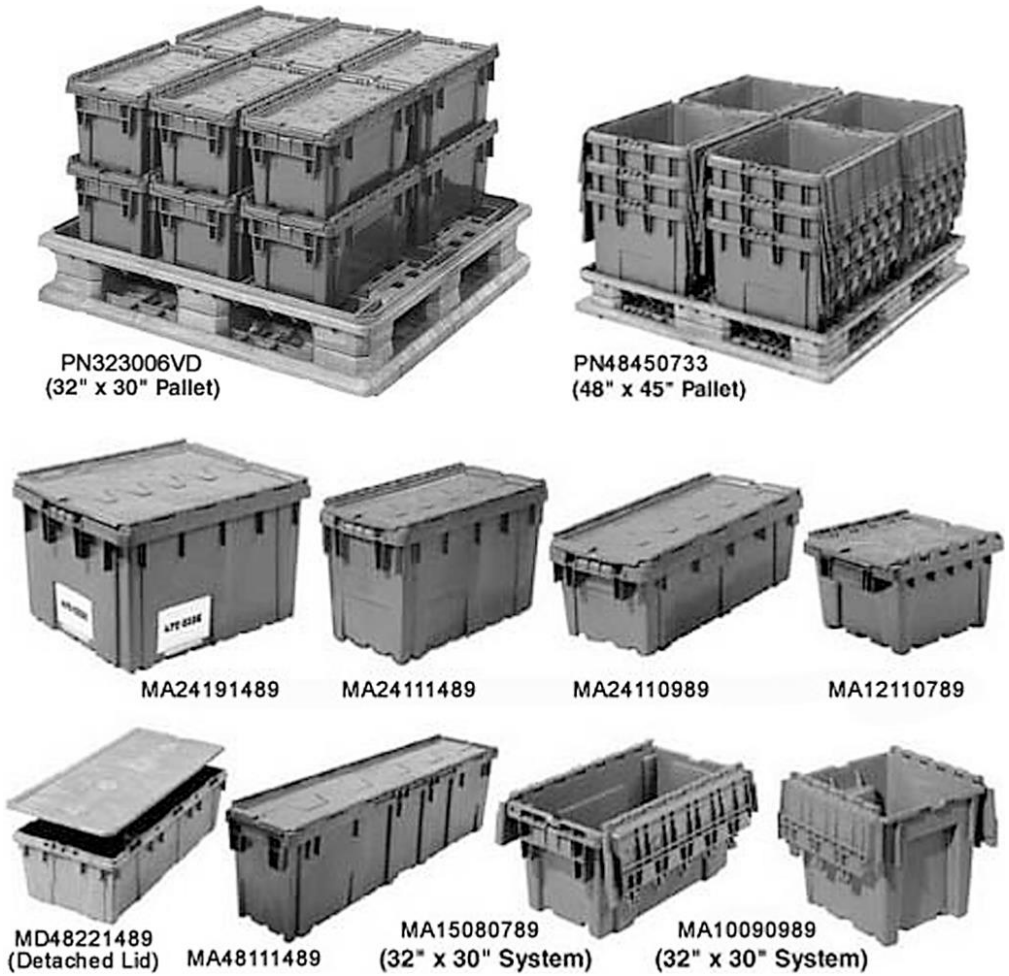
# Unit Load


Carton
12" × 8" × 10"
Wood Pallet (4-way)
32" × 48" × 5.5"
Pallet Load
32" × 48" × 45.5"
16 cartons per layer
4 layers per pallet
Block Stack Storage
Wall
32"
32"
32"
32"
50"
50"
50"
50"
50"
48" + 2" Clearance
Trailer
8' – 6" I.D.
32"
48"
32"
48"

# Returnable Containers (Continue on Unit Load)

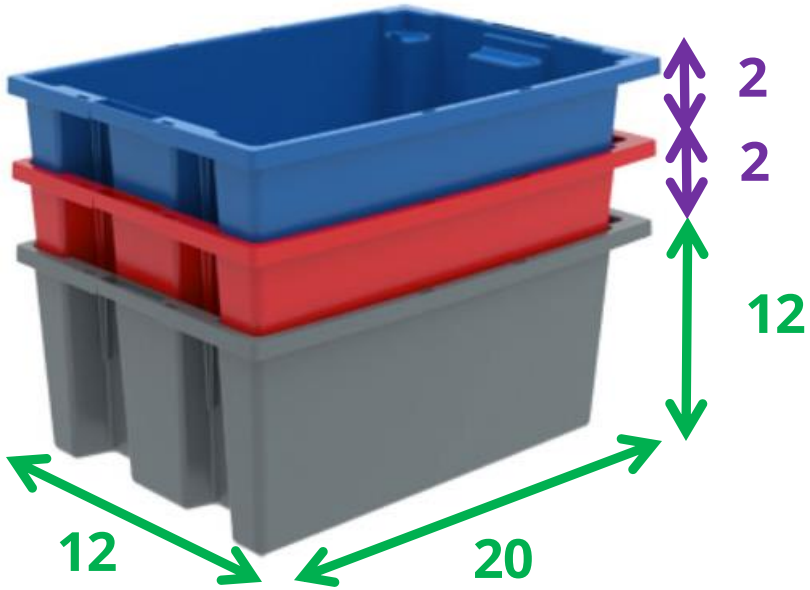
Returnable containers with good stacking and nesting features are preferred

Examples: Plastic totes, Metal racks, Cardboard boxes



# Returnable Containers (Continue on Unit Load)

- Returnable containers of following dimensions are available.
- Inside dimensions (L x W x H): 18” x 11” x 11”
- Outside dimensions: 20” x 12” x 12”
- Each nested empty containers: 20” x 12” x 2”



# Returnable Containers

- A trailer with inside dimensions of 240” x 120” x 120” is used to transport containers. Containers are not palletized. No clearance is needed between containers or between containers and wall of the trailer

- Determine *container space utilization*

Inside: 18” x 11” x 11”
Outside: 20” x 12” x 12”
Nested: 20” x 12” x 2”

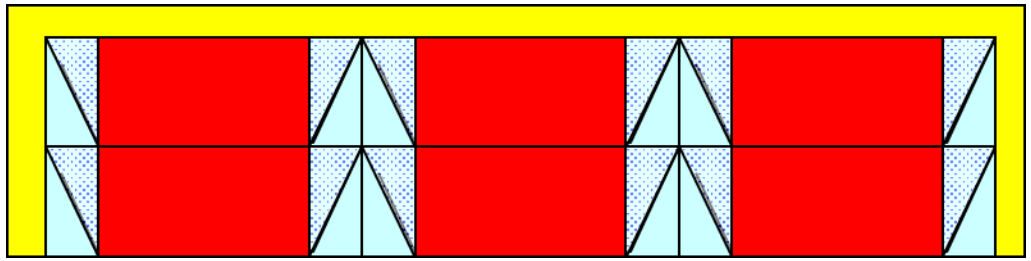


# Returnable Containers

- A trailer with inside dimensions of 240” x 120” x 120” is used to transport containers. Containers are not palletized. No clearance is needed between containers or between containers and wall of the trailer

- Determine *storage space efficiency*

  Inside: 18” x 11” x 11”
  Outside: 20” x 12” x 12”
  Nested: 20” x 12” x 2”

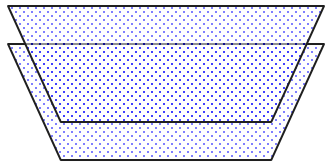


# Returnable Containers

- A trailer with inside dimensions of 240” x 120” x 120” is used to transport containers. Containers are not palletized. No clearance is needed between containers or between containers and wall of the trailer

- Determine *container resting ratio*

  Inside: 18” x 11” x 11”
  Outside: 20” x 12” x 12”
  Nested: 20” x 12” x 2”



- Ratio is 6:1

i.e. six nested containers take up same space as 1 closed container

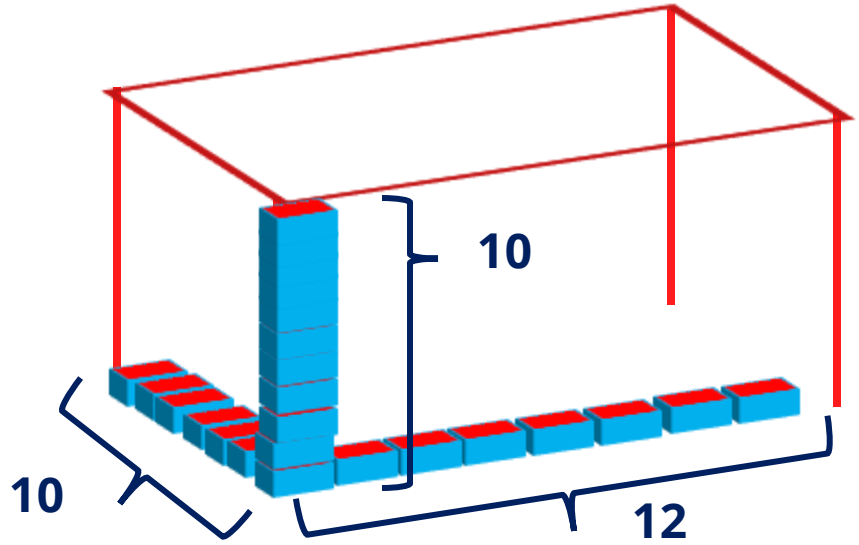
# Returnable Containers

- A trailer with inside dimensions of 240” x 120” x 120” is used to transport containers. Containers are not palletized. No clearance is needed between containers or between containers and wall of the trailer

3. Determine trailer space utilization

- (when all containers are stacked vertically in only one orientation)
- Length: 240 / 20 = 12 containers side by side
- Width: 120 / 12 = 10 containers side by side
- Height: 120 / 12 = 10 containers vertically
- Total of 12 x 10 x 10 containers in a trailer

Inside: 18” x 11” x 11”
Outside: 20” x 12” x 12”
Nested: 20” x 12” x 2”



# Returnable Containers

- A trailer with inside dimensions of 240” x 120” x 120” is used to transport containers. Containers are not palletized. No clearance is needed between containers or between containers and wall of the trailer

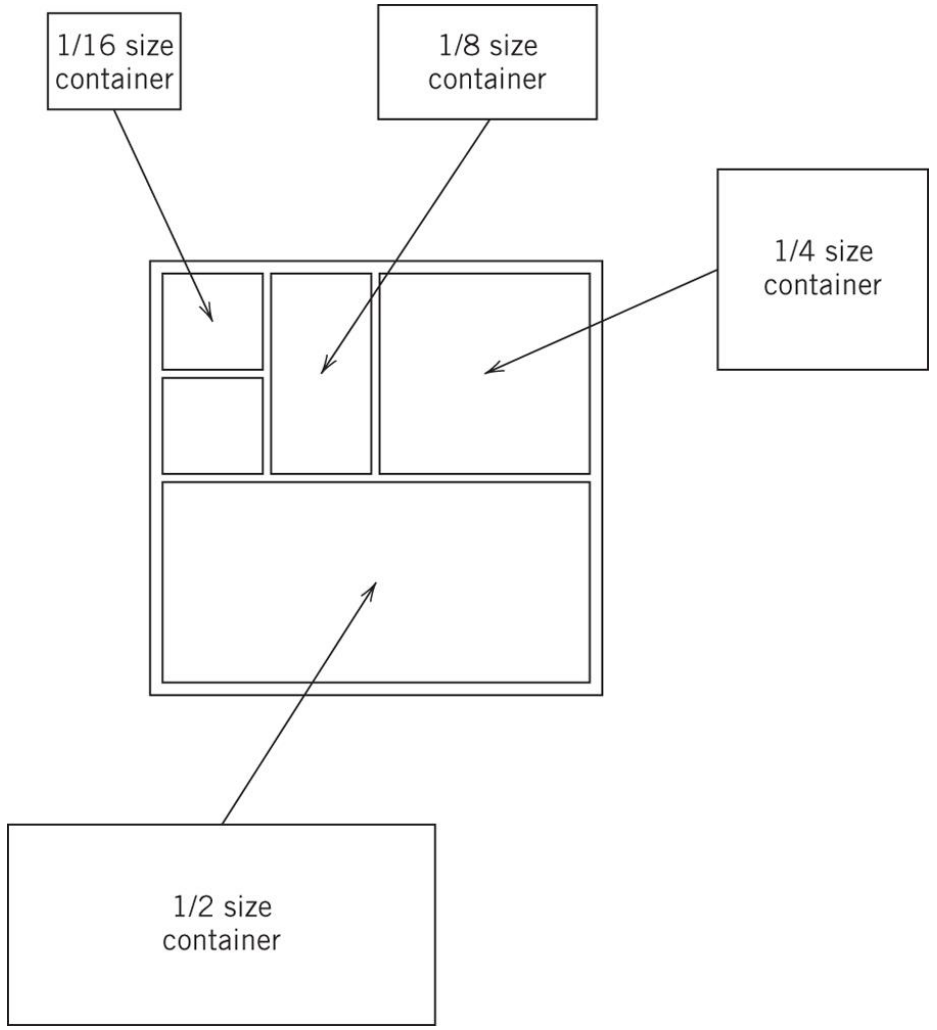
Inside: 18” x 11” x 11”
Outside: 20” x 12” x 12”
Nested: 20” x 12” x 2”

3. Determine trailer return ratio

- When full, 120/12 = 10 containers can be stacked vertically
- When empty, 1 + (120 – 12) / 2 = 55 containers can be stacked vertically
- Width: 240 / 20 = 12 containers side by side
- Length: 120 / 12 = 10 containers side by side
- Total of 12 x 10 x 55 = 6600 empty containers in a trailer
- 6600 empty containers / 1200 full containers = 5.5

# Returnable Containers

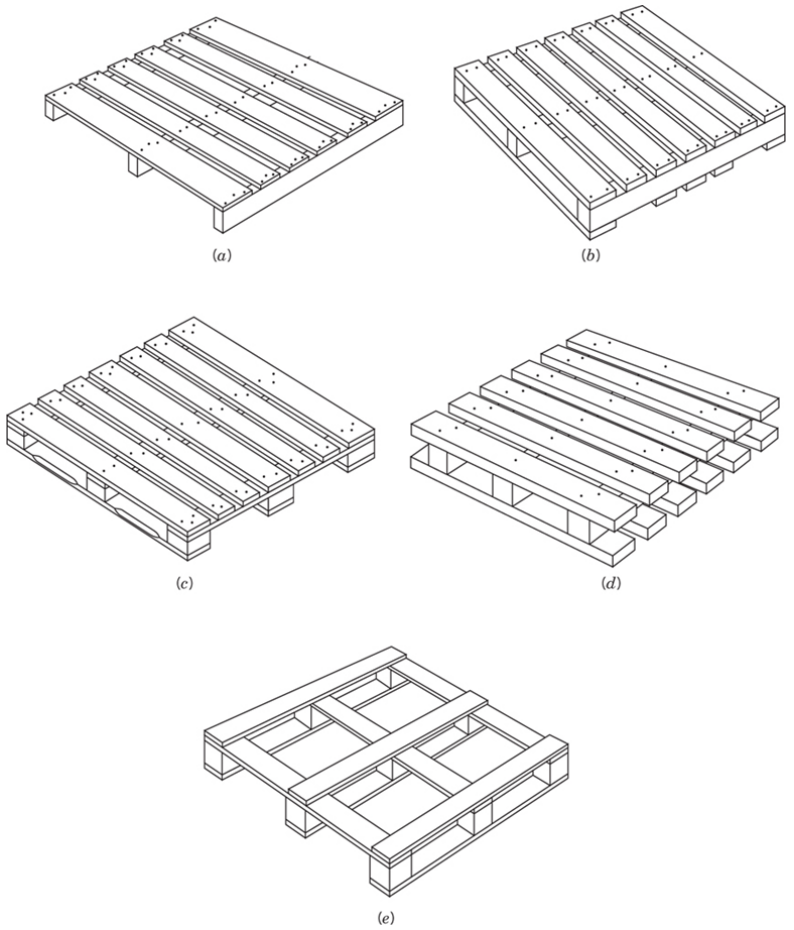
- Size progression is one of important considerations
- This way, trailer can be fully loaded, achieving 100% utilization



# Pallets

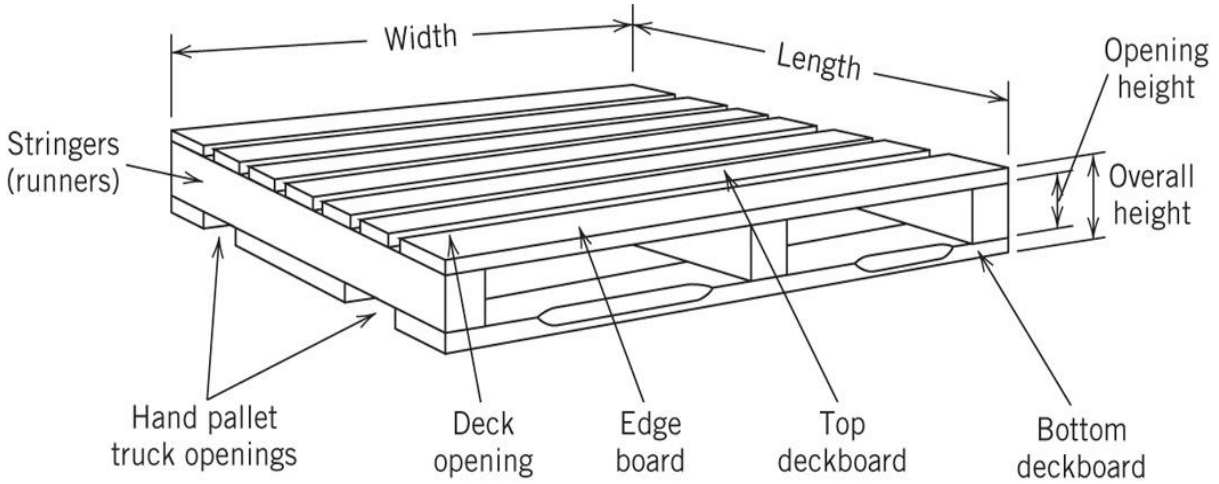
- A pallet is another common method of containing unit load





# Pallets




Stringer Design



| Material | Base Weight | Durability | Reparability | Environmental Impact | Typical Applications |
|---|---|---|---|---|---|
| Wood | 15-112 lb | Medium | High | Biodegradable Recyclable | Wide general use, grocery, automotive, durable goods, hardware |
| Pressed Wood Fiber | 30-42 lb | Medium | Low | Recyclable Can be burned without fuel residues | Bulk bags, order-picking, printing, building material |
| Corrugated Fiberboard | 8-12 lb | Low | Low | Biodegradable Recyclable | Export shipping, one-way shipping, grocery, lightweight paper products, industrial parts |
| Plastic | 35-75 lb | High | Medium | Recyclable | Captive or closed loop systems, FDA, USDA applications, AS/RS, automotive |
| Metal | 32-100 lb | High | Medium | Recyclable | Captive or closed loop systems, FDA, USDA applications, AS/RS, military, heavy equipment, aerospace |

# Material Handling Equipment (MHE)

- 1. Containers and unitizing equipment,
- 2. Material transportation equipment,
- 3. Storage and retrieval equipment,
- 4. Automatic identification and communication equipment

Refer to Appendix 5 Material Handling Equipment in Thompson et al. for details on material handling equipment

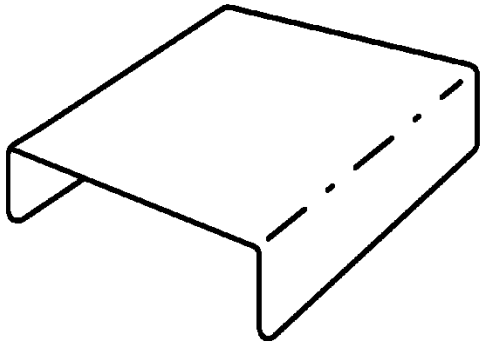
# 1. Containers & Unitizing Equipment

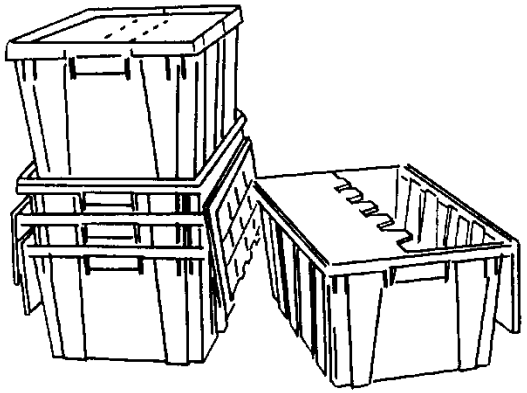
A. Containers



Pallets

Skids

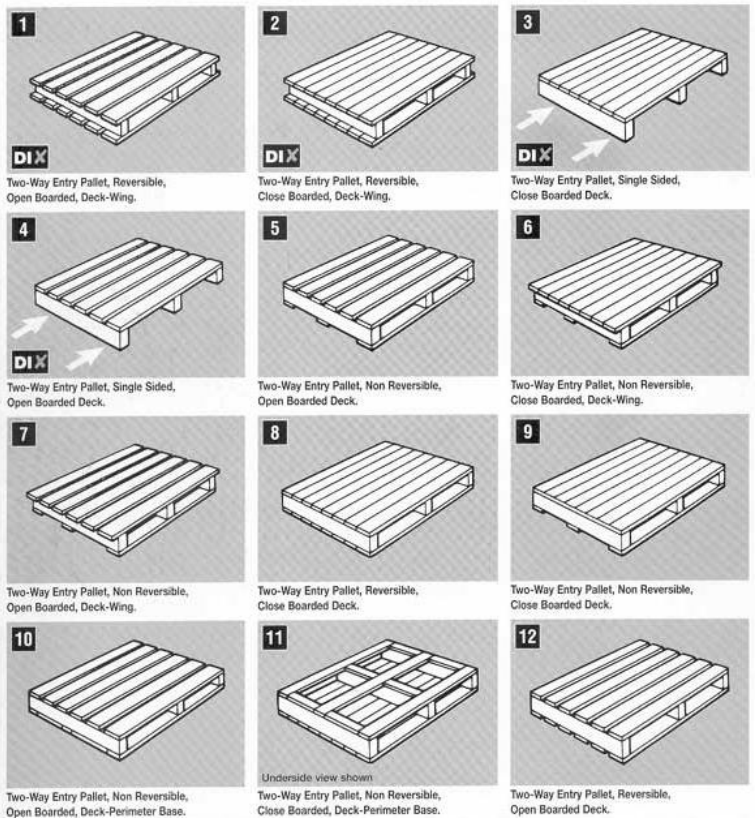




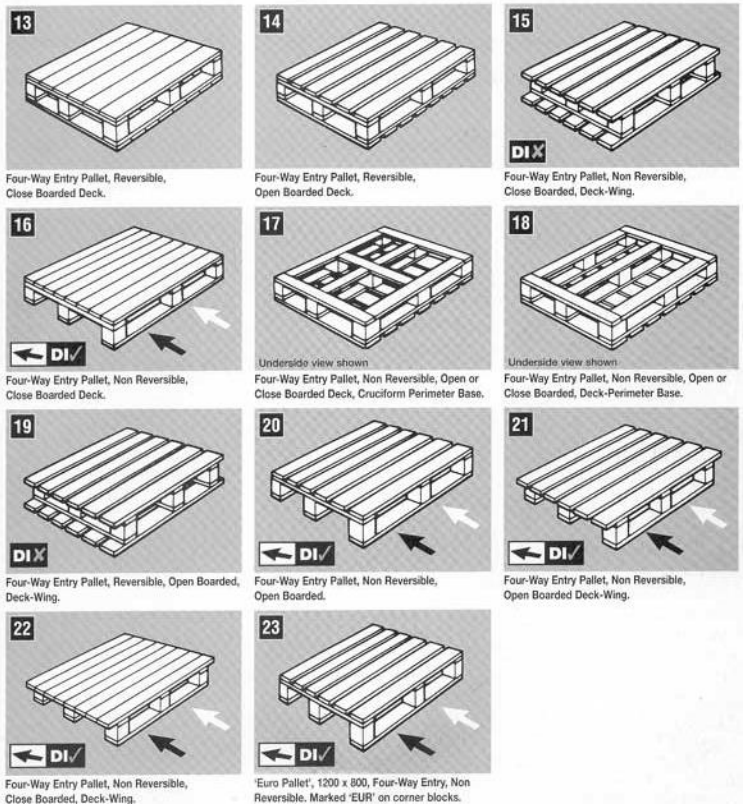
Tote pans

# Pallets

- Two-way

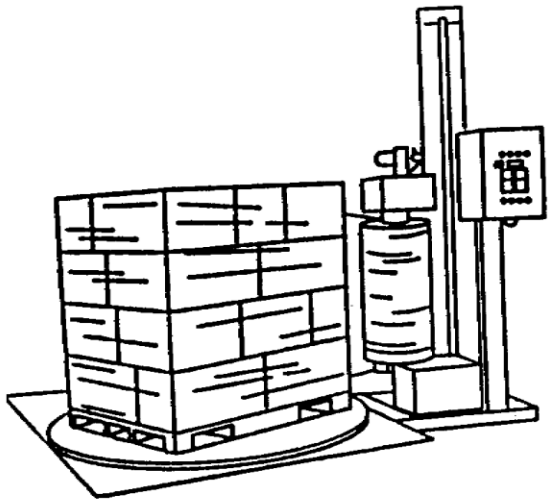



- Four-way



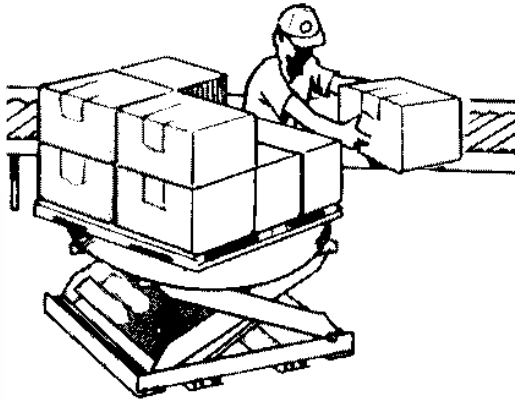
# 1. Containers & Unitizing Equipment

## B. Unitizers





Stretchwrap





Palletizers

# Palletizer

- A machine which provides automatic means for stacking cases of goods or products onto a pallet











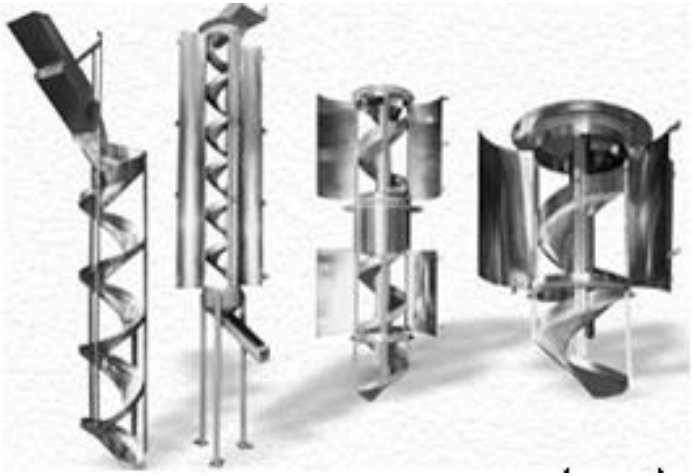
# 2. Material Transportation Equipment

- A. Conveyors

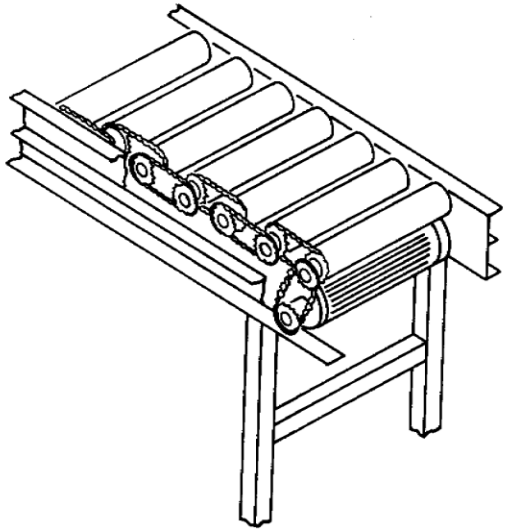
**Conveyor:** a fixed-path materials handling device



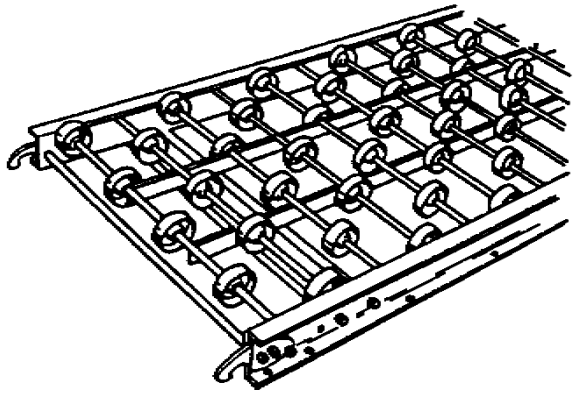
**Chute Conveyor:** A conveyor used to transport material from a higher level to a lower level via gravity chutes.



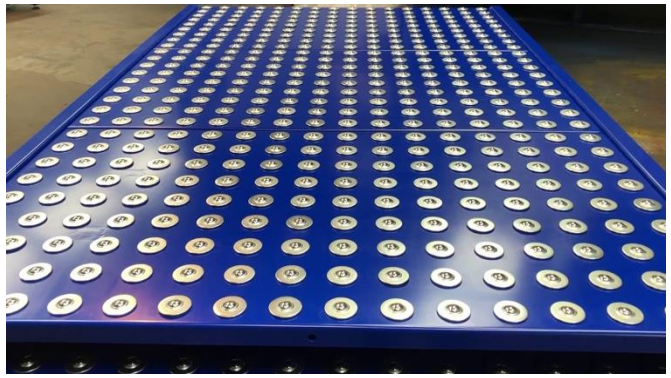
# 2.A. Conveyors (Cont.)



Roller conveyor



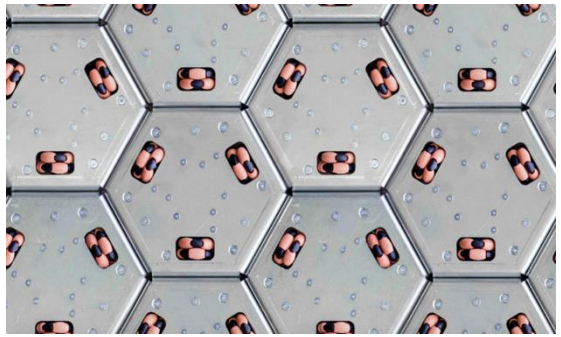
Wheel-conveyor



Ball conveyor

**Slat Conveyor:** A conveyor constructed of one or more strands of chain to which steel or thermoplastic slats are attached to convey a variety of parts or packages.

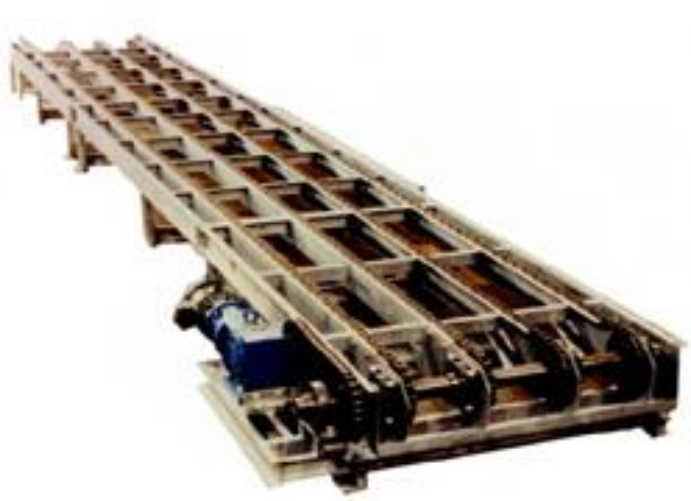




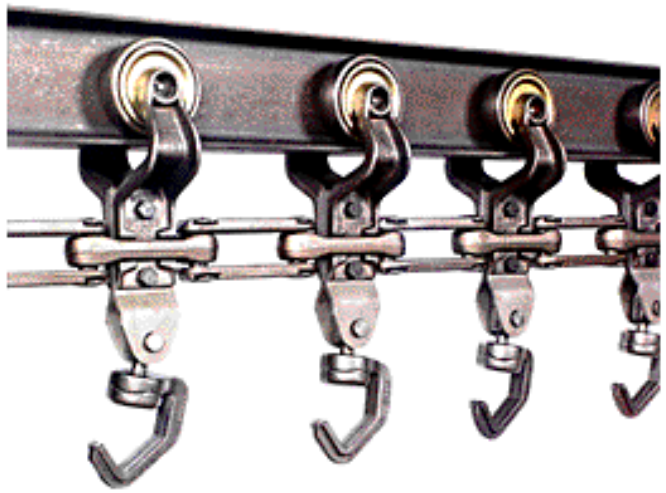
Modular conveyor

# 2.A. Conveyors (Cont.)

**Chain Conveyor:** A conveyor that is typically an overhead device with carriers transported via chains.

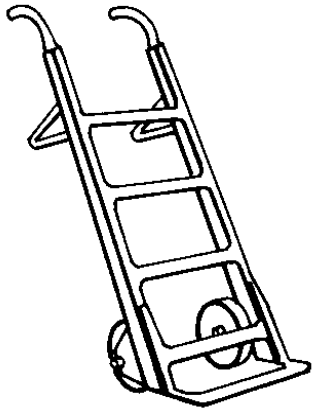


**Trolley Conveyor:** A device that transports loads in a series of trolleys attached to a chain and supported by or within an overhead track.

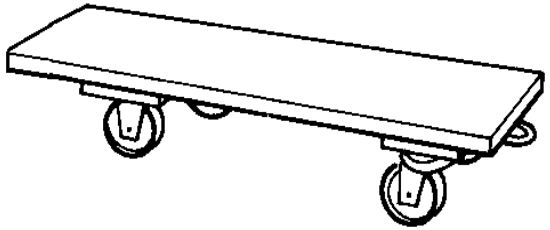


# 2.B.Industrial Vehicles (Cont.)

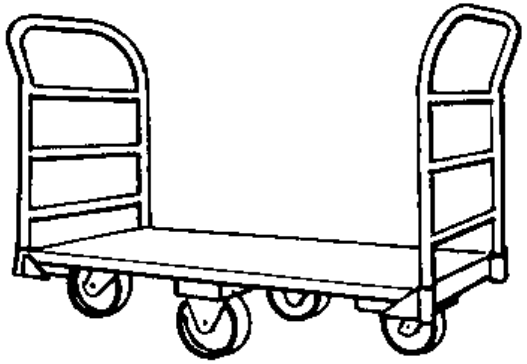
- Hand Trucks



Two-wheeled hand truck



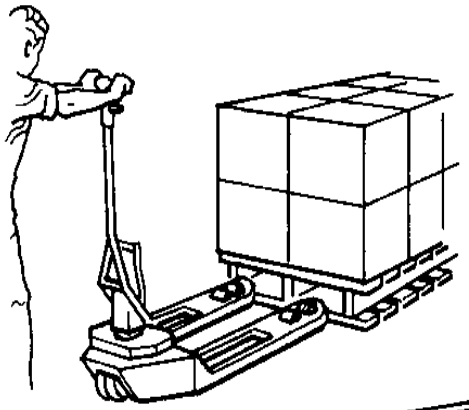
Dolly

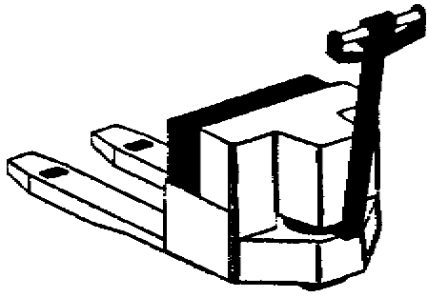


Floor hand truck

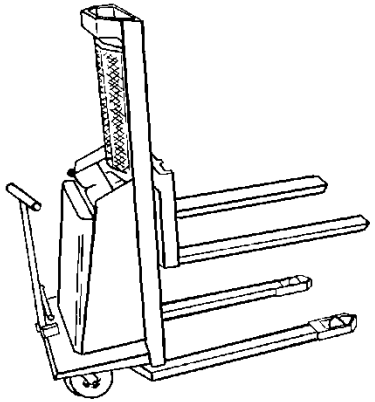
# 2.B.Industrial Vehicles (Cont.)

Manual Pallet Jack

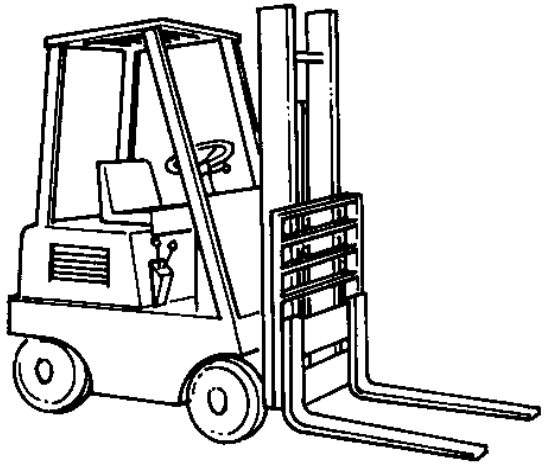




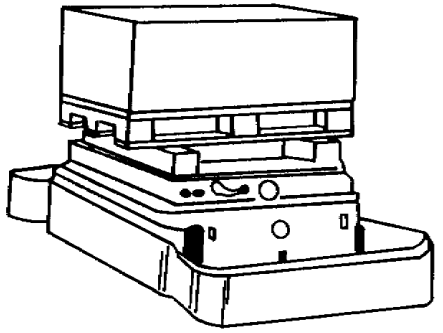
Powered Pallet Jack



Manual Walkie Stacker

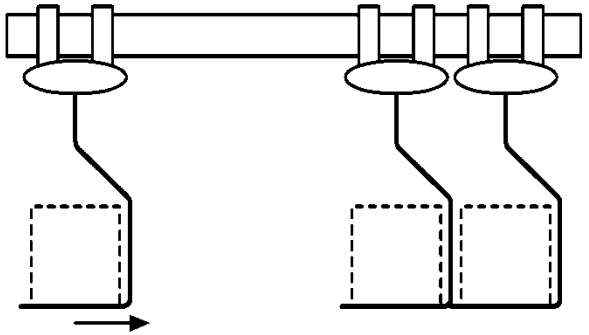


Counterbalanced (CB) Lift Truck

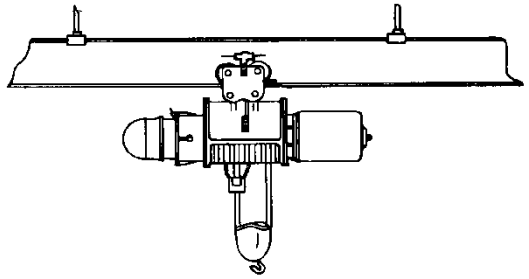


Unit load AGV

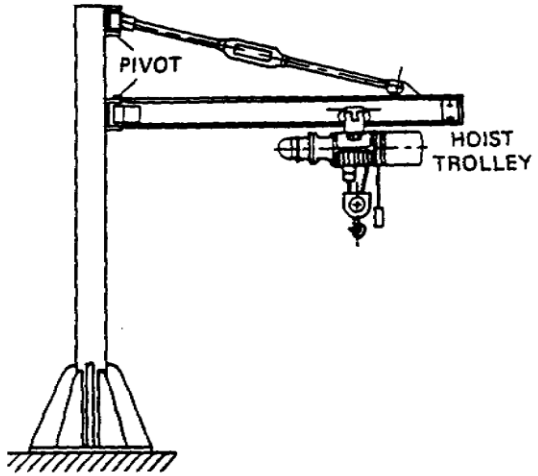
# 2.C. Monorails, Hoists, and Cranes



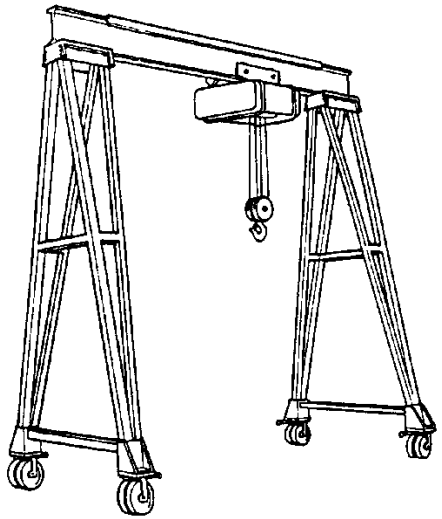
Monorail



Hoist



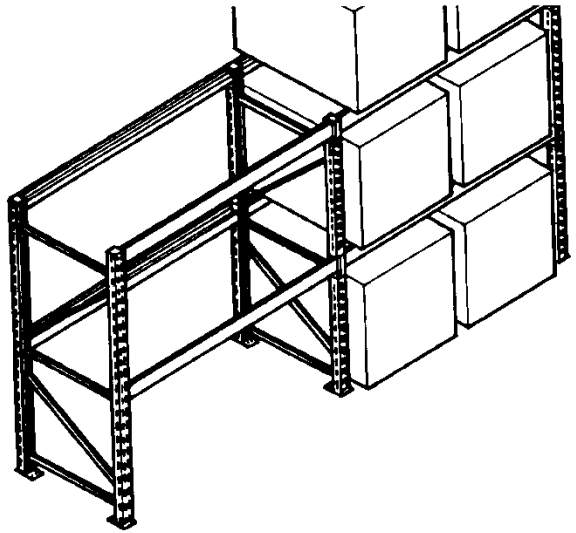

Jib Crane



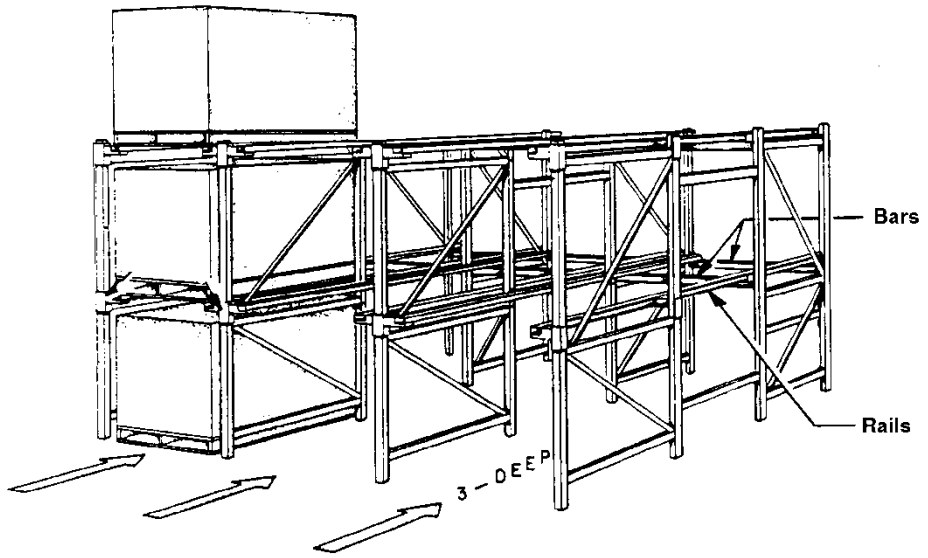
Gantry Crane

# 3. Storage and Retrieval Equipment

- 3.1. Unit load storage and retrieval

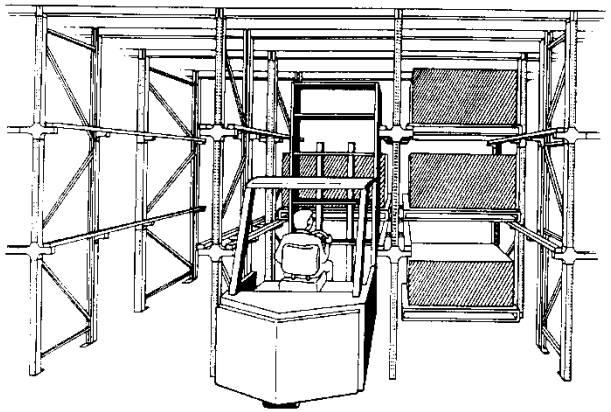


Single deep selective rack

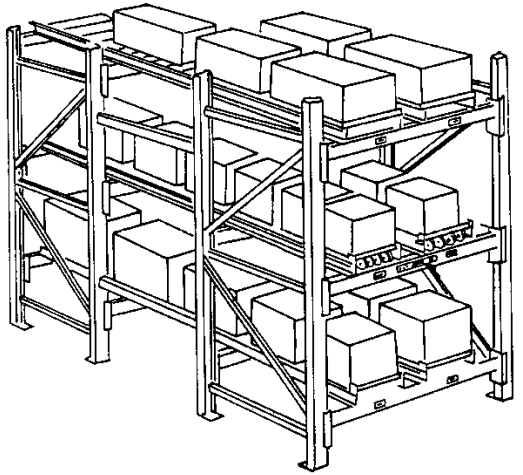



Drive-in rack

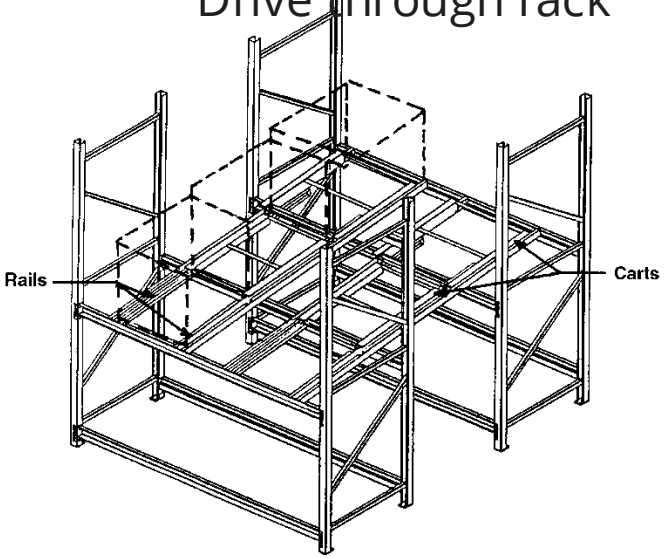
# 3.1 Unit Load Storage and Retrieval



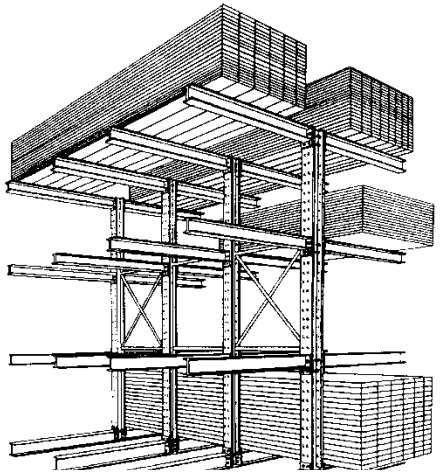
Drive through rack



Pallet flow through rack



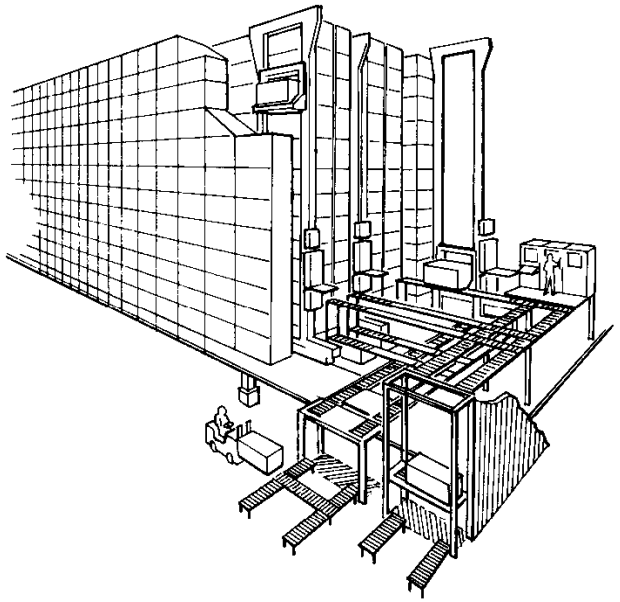

Push back rack



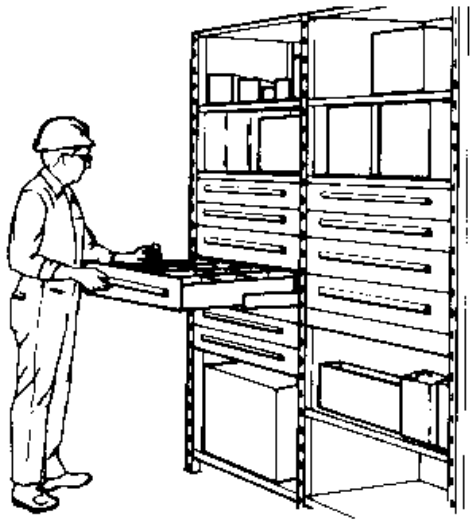
Cantilever rack

# 3.2 Unit Load Retrieval Equipment

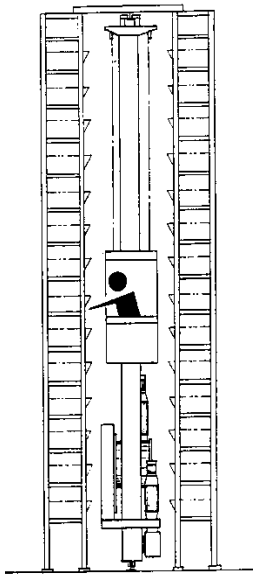
- Walkie stacker
- Counterbalanced lift truck
- Narrow aisle vehicles
- Automated storage/retrieval systems



# B. Small Load Storage and Retrieval



Drawers, bins



Person-aboard AS/RS



Carousel

# Warehouse

## Single Deep Selective Rack

Single-deep pallet racking provides access to each pallet stored in the rack. The major disadvantage is that the racks require significant floor space for suitable aisles.





## Block Stacking

Block stacking refers to unit loads stacked on top of each other and stored on the warehouse floor in lanes or blocks. The pallets are stacked to a specific height based on a number of criteria such as pallet condition, weight of the load, height clearance and the capability of the warehouse forklifts.



# 4. Automatic Identification and Communication Equipment

## Automatic identification and recognition

1. Bar coding
2. Optical character recognition
3. Radio frequency tag
4. Magnetic stripe
5. Machine vision

## Automatic communication

1. Radio frequency data terminal
2. Voice headset
3. Light and computer aides
4. Smart card

# Designing Material Handling Systems

Steps

- Define the objectives and scope for the system
- Analyze the requirements for moving, storing, protecting, and controlling material
- Generate alternative designs for meeting material handling system requirements
- Evaluate alternative designs
- Select the preferred design
- Implement the design, periodic audits system performance and make potential modifications

# Material Handling Planning Chart

Gather information pertaining to a specific material handling problem and provide a preliminary examination of the alternative solutions.

**MATERIAL HANDLING PLANNING CHART**

Company A.R.C. Inc, Prepared by I.A. Layout Alternative 1

Product Air Speed Control Value Date Sheet 1 of

| Step No. | O | T | S | I | Description | Oper No. | Dept. | Cont. Type | Size | Wt. | Qty. Per Cont. | Freq | Dist | Method of Handling |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | X | | Bar stock in storage (2200) | | Stores. | | | | | | | |
| 2 | | X | | | Profit stores to saw dept. | | | LDDSE (FK.TRK) | 2.5" × 3.5 × 16" | 5 lb | to bars | 3 times daily | 16 ft | Fork lift |
| 3 | | | X | | Store in saw department | | Saw | | | | | | | |
| 4 | X | | | | Cut to length | 0101 | Saw | | | | | | | |
| 5 | | X | | | From saw to grinding | | | TOTE pan | 15" × 12" × 7" | 30 lb | 30 | Twice daily | 10 ft | Platform hand truck |
| 6 | | | X | | Store in grinding | | Grinding | | | | | | | |
| 7 | X | | | | Grind to length | 0201 | Grinding | | | | | | | |
| 8 | | | X | | | | | TOTE pan | 15" × 12" × 7" | 30 lb | 30 | Twice daily | 13 ft | Platform hand truck |
| 9 | | | | X | Store in deburring | | Deburring | | | | | | | |
| 10 | X | | | | Deburr | 0301 | Deburring | | | | | | | |
| 11 | | X | | | From deburring to dr. Prs | | | TOTE pan | 15" × 12" × 7" | 30 lb | 30 | Twice daily | 16 ft | Platform hand truck |
| 12 | | | X | | Store in drill press | | Drill Press | | | | | | | |
| 13 | X | | | | Dr. CD holes tap. rean, dsk | 0401 | Drill Press | | | | | | | |
| 14 | | X | | | From dr. press to tur. lathe | | | TOTE pan | 15" × 12" × 7" | 30 lb | 30 | Twice daily | 33 ft | Platform hand truck |

185

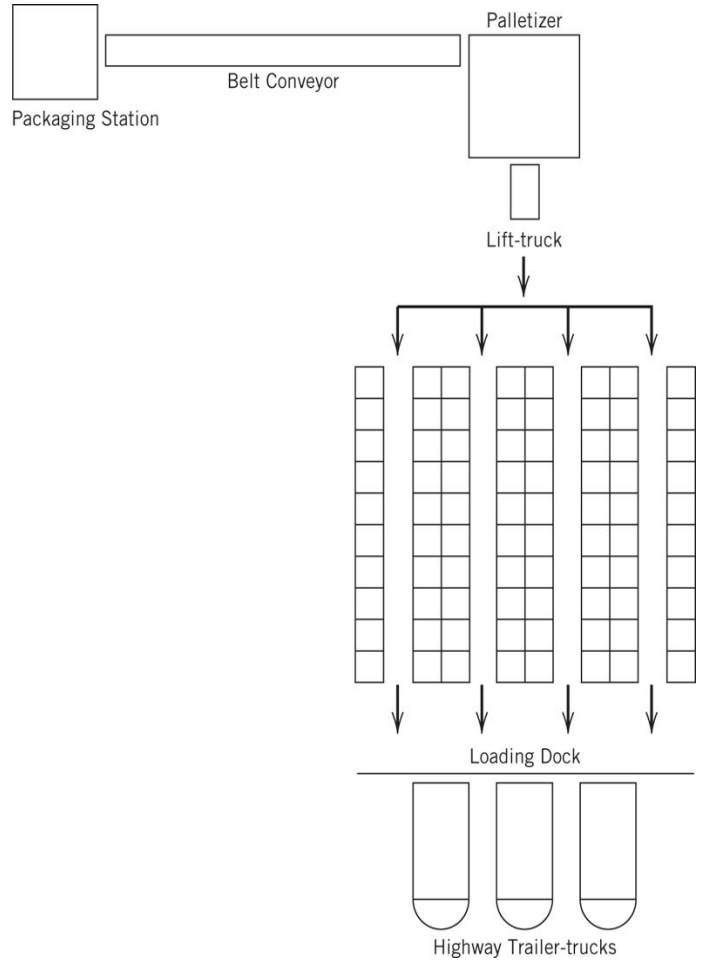
**Figure 5.2** Material handling planning chart for an air flow regulator. Key: Operation—O. Transportation—T. Storage—S. Inspection—I.

# Material Handling Planning Chart

- Each operation (O), transport (T), storage (S), and inspection (I) activity are listed in the chart.
- A transport activity for O, T and S should be recorded, in the format of “From __ to __”. – which answers where the flow moves.
- A final decision on the movement cannot be made from this chart, since the type of unit load and its size are not determined.

# Unit Load Design

- A material handling system consists of packaging, palletizing, storing and shipping operations.
- Operations include:
  - Finished goods are packaged using closed-top cardboard cartons
  - Cartons are transported to a palletizer via a belt conveyor.
  - Pallet loads are formed using a *mechanical palletizer*
  - Full pallet loads are then stored in finished-goods warehouse using *powerlift truc*k (warehouse use only).
  - Upon receipt of customer orders, full pallet loads are retrieved from warehouse by *powered lift truck*. (shipping dock operations only)
  - Retrieved pallet loads are then loaded on highway *trailer trucks* for delivery to customers.



# Unit Load Design

**For simplified illustration, we assume that only one product category is used.**

**Following alternatives are considered:**

1. Carton size
   - 12” x 10” x 10”
   - 10” x 8” x 8”
2. Pallets
   - 40” x 48” (two-way)
   - 48” x 40” (two-way)
   - 36” x 36” (four-way)
3. Palletizer
   - Max Load height 70”
4. Lift truck (warehouse operation)
   - Counterbalance lift truck with simplex mast, 6000lb at 24” load center, max fork height 106”
   - Counterbalance lift truck with duplex mast, 6000lb at 24” load center, max fork height 130”
   - Narrow isle high-rise lift truck 6000lb at 24” load center, max fork height 20”
5. Warehouse storage
   - Single-deep pallet rack design with 20” clear height
   - Block stacking design with 20” clear height
6. Lift truck (shipping dock operations)
   - Counterbalance lift truck with duplex mast, 6000lb at 24” load center, max fork height 106”
7. Highway Trailer Truck
   - 7’6” x 60’ x 11’ (inside dimensions)